Machine Learning Capstone Project: Used Cars Price Prediction Low-Code Final Submission Report

MIT Applied AI and Data Science Program

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Executive Summary

Cars4U faces a predominant business stipulation. This business stipulation pertains to the Indian used-car market, which is highly unstable, fragmented, and nonlinear, leading to inaccurate used-car prices for both buyers and sellers. To address this business requirement, the project built a machine-learning-driven pricing engine, selecting a Random Forest Regression model as the final solution. The model delivers accuracy, stability, and generalization, making it the most reliable model for Cars4U's automated pricing system.

The analysis of the Cars4U used-car pricing project demonstrated that used-car prices vary significantly by brand, Indian city, fuel type, and ownership history. The provided dataset, while rich, has considerable missing values and right-skewed distributions in numerical variables. Following thorough preprocessing and feature engineering, clear economic patterns are evident in the dataset. Higher engine power, larger engine size, and new-car price lead to higher used-car prices. In contrast, age, mileage, and kilometers driven are associated with lower used-car prices. Comparisons across linear and geometric models could not capture the intricacy of the Indian used-car market. Tree-based nonlinear models performed much better, where the Random Forest Regression model achieved the best results.

The final Random Forest Regression model provides Cars4U with a scalable, interpretable, and business-aligned pricing engine. The model is competent in reducing negotiation friction, refining inventory decision-making, and improving customer trust.

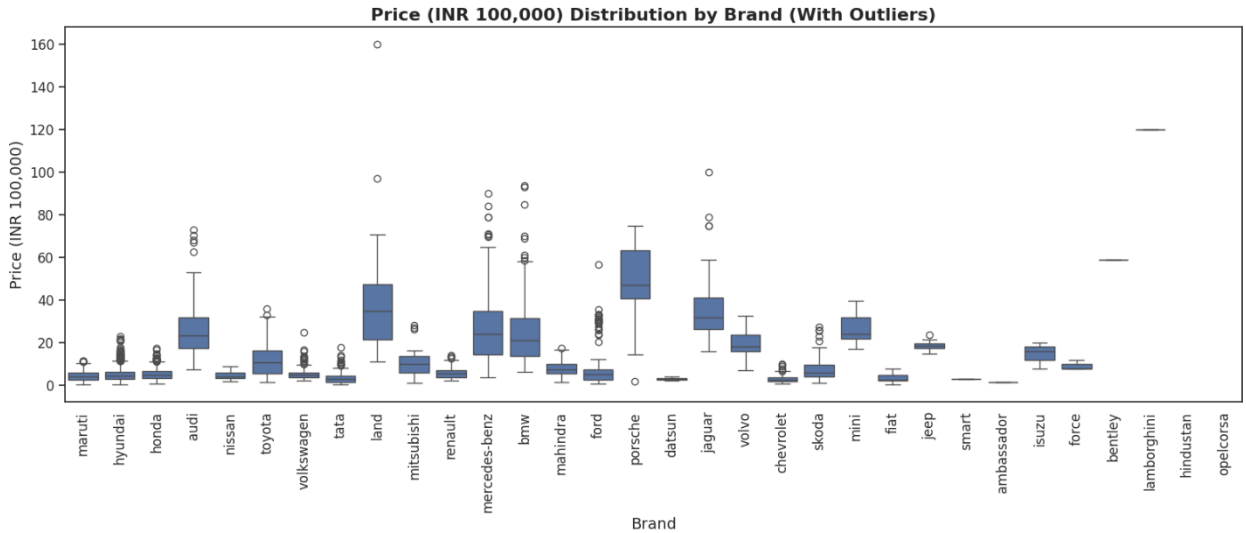
Most important findings from the analysis done in Milestone

The following statements are the most important findings from the analysis done in the previous Milestone report. Foremost, the Indian used-car market exhibits significant nonlinearity and high variability. Through careful exploration, the Cars4U dataset indicates an exceedingly diverse market. Within this market, there are formidable differences across brands, cities, fuel types, and ownership histories. In particular, the market is not consistent, as pricing valuations vary predominantly through dominant nonlinear relationships. This insight directly supports the rationale for later preferring nonlinear models.

The data visualizations below depict the nonlinearity, variability, and diversity of the Indian used-car market. The first two data visualizations show the critical differences between Brand and City with the target Price variable. When analyzed together, the data show strong pricing variability and nonlinearity, as well as market

inconsistencies across key categories. The Fuel Types and Owner Type bar charts illustrate the characteristics of the Indian used-car market.

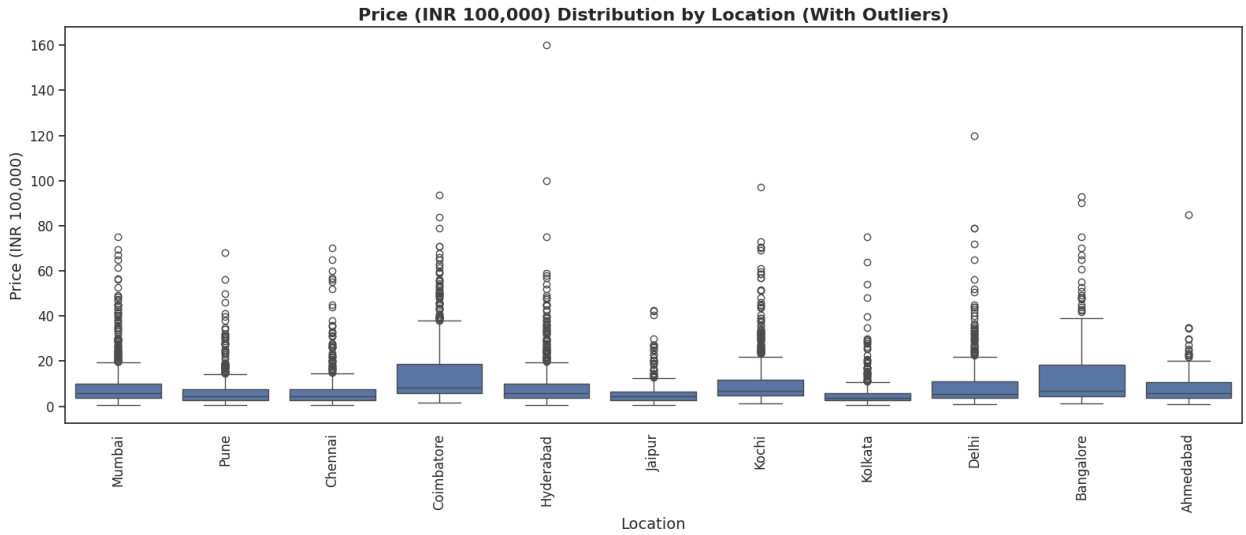
Figure 1. Price (INR 100,000) Distribution by Brand (With Outliers) Boxplot



Observations and Insights on Price (INR 100,000) Distribution by Brand (With Outliers) Boxplot

The Price distribution by brand, with outliers shown in the boxplot, suggests the following insights. First, luxury brands, including Mercedes-Benz, BMW, Audi, Jaguar, Porsche, and Bentley, have the highest medians, the most variability, and the most outliers. This indicates the presence of premium models with a wider price spread. More mainstream brands, including Maruti, Hyundai, Honda, Toyota, Ford, Mahindra, and Tata, cluster at lower- to mid-range price points, where pricing is more predictable, and there are fewer outliers. Brands, including Land Rover and Mitsubishi, have higher medians and wider spreads, suggesting a mix of rugged SUVs and premium variations. Sparse, infrequent brands, including Smart, Ambassador, Isuzu, Force, and Opel Corsa, have narrow boxplots due to very small sample sizes, making the price patterns less reliable. In general, outliers are most often found in luxury and upper-tier brands, which indicate the highest-value used-car vehicles. Here, we can also see that the market is clearly segmented: high-volume, price-sensitive brands are tightly clustered, while luxury brands compete in a fragmented, high-variance pricing landscape. Brands with narrow distribution, such as Maruti and Hyundai, rely on standardized, value-driven offerings, which limit price ranges.

Figure 2. Price (INR 100,000) Distribution by Location (With Outliers) Boxplot

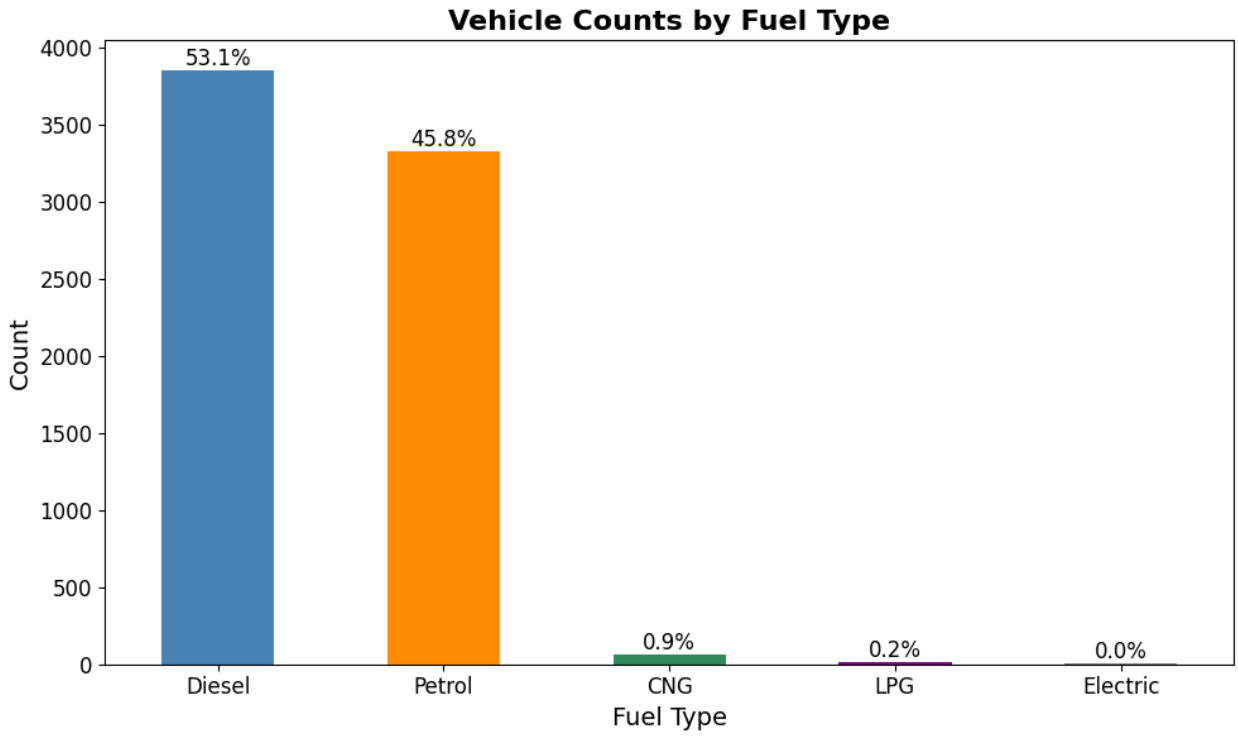


Observations and Insights on Price (INR 100,000) Distribution by Location (With Outliers) Boxplot

The Price Distribution by Location boxplot with outliers shows extreme outliers and significant price variation across Indian cities. There is evidence that several markets show widespread, numerous high-value outliers. Coimbatore and Bangalore have the broadest interquartile ranges and the largest clusters of extremely high-priced points, suggesting amalgamated markets with both budget and premium segments. On the other hand, cities including Jaipur, Kochi, and Ahmedabad indicate firmer distributions, indicating more homogeneous pricing and far fewer luxury-tier vehicles. Likewise, cities such as Mumbai, Delhi, Bangalore, and Coimbatore show numerous outliers, indicating that luxury used-car brands, high-spec trims, and newer used vehicles are more common in these cities and fall above common market ranges. These outliers underscore that location is related to other predictors (e.g., brand, power, and new price) and can shape price variability. The boxplot indicates that location affects price levels, with some cities exhibiting higher and more volatile price ranges than others. Some insights include that the market is geographically tiered, with key, economically significant regions commanding higher price points, suggesting stronger demand for newer or premium vehicles. Also, Indian cities such as Mumbai, Delhi, and Bangalore show greater variability and many outliers, indicating diverse

buyer segments ranging from budget to luxury used-car buyers. Lastly, lower-variance cities, such as Ahmedabad, Jaipur, and Kolkata, exhibit more uniform market behavior, as reflected primarily in mid-range or value-oriented vehicles.

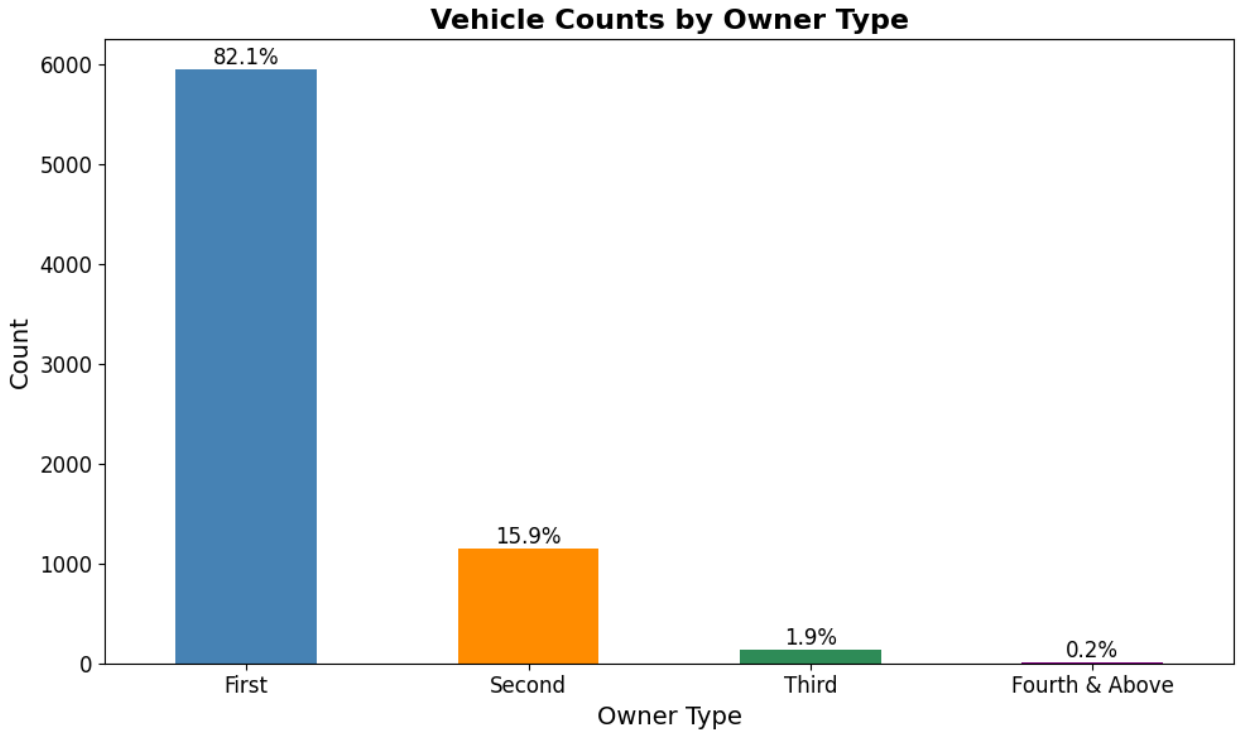
Figure 3. Vehicle Counts by Fuel Type Bar chart



Observations and Insights on Vehicle Counts by Fuel Type Bar chart

The Fuel Type bar plot shows 5 distinct Fuel Types among the used cars in the Cars4U inventory—the majority run on diesel or petrol. About 53.1% of all used cars run on diesel, and about 45.8% run on petrol. This dominance of diesel and petrol is expected in the actual Indian used-car market, since diesel cars were prominent during 2010-2016 due to fuel price fluctuations and torque advantages. Along the same lines, used petrol-powered cars are usually small hatchbacks and compact sedans. As for used cars that run on CNG or LPG, or are electric, they are very uncommon. The market is largely reliant on traditional combustion engines, with alternative fuels much less common. The Diesel fuel type leads indicate stronger adoption in segments where torque, efficiency, or long-distance use is more important. The very low number of electric used-cars indicates early adoption, most likely suppressed by infrastructure, cost, or consumer awareness.

Figure 4. Vehicle Counts by Owner Type Bar chart



Observations and Insights on Vehicle Counts by Owner Type Bar chart

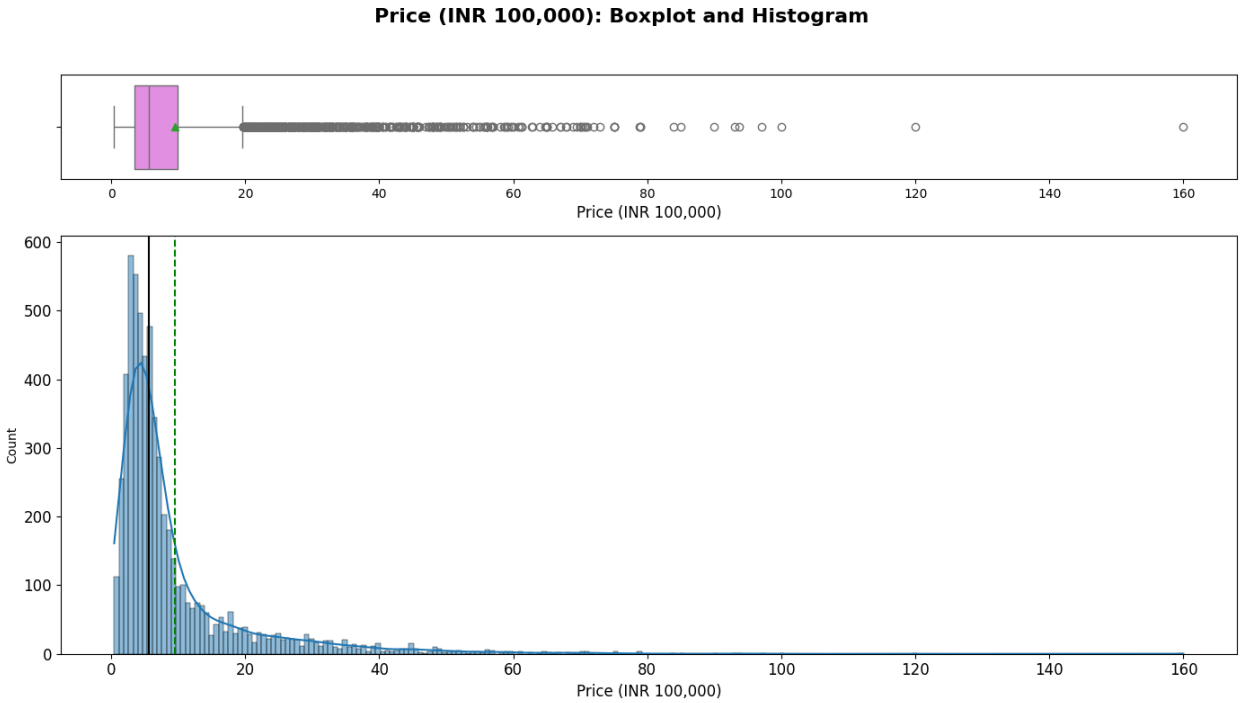
The Owner Type bar plot indicates a clear dominance of first-owner used cars, which aligns with realistic customer behavior. As 82.1% of used cars are classified as first-owner type, this means they are more robust, easier to sell, command higher resale prices, are recognized for better maintenance, and are associated with cleaner service histories. Second-owner used cars still account for a significant minority, about 15.9%. These used cars are typically still less costly than first-owner used cars and often have higher mileage.

Second, essential numerical independent variables, including engine displacement, power, new prices, and used-car prices, exhibited right-skewness due to the presence of luxury vehicles. Essential categorical independent variables, such as Fuel Type and Owner Type, aligned with clear Indian used-car market characteristics, as diesel cars and first-owner vehicles comprised most of the dataset. There were significant, but manageable, data quality issues. The exploratory data analysis revealed significant missing values in the New Price, Power, Engine, Mileage, Seats, and target Price variables. To clarify, there were over 6,247 missing values for the New Price variable, where there were similar noteworthy gaps in the remaining Power,

Engine, Mileage, Seas, and target Price variables. Due to these data quality issues, the dataset required thorough preprocessing. These methods included imputation, outlier consideration, and feature engineering. Regardless of these issues, the dataset still provided powerful signals for modeling after it was cleaned and properly engineered. If these preprocessing methods had not been implemented, no model could reliably generalize from the dataset.

The data visualizations below depict the skewness, missing data, and data quality issues found in the provided dataset. The first two data visualizations show the significant right-skewness of key numerical variables, such as the target Price variable and Kilometers Driven, before applying a log transformation. The third, fourth, fifth, and sixth visualizations show significant right skewness in Power, New Price, Engine, and Mileage, respectively. When analyzed together, the graphs indicate that the dataset is mostly composed of affordable, lower-mileage used cars, with fewer premium or luxury vehicles and high-mileage used cars. The final Missing Values bar chart shows the extent of missingness across the Mileage, Engine, Power, Seats, New price, and Price variables.

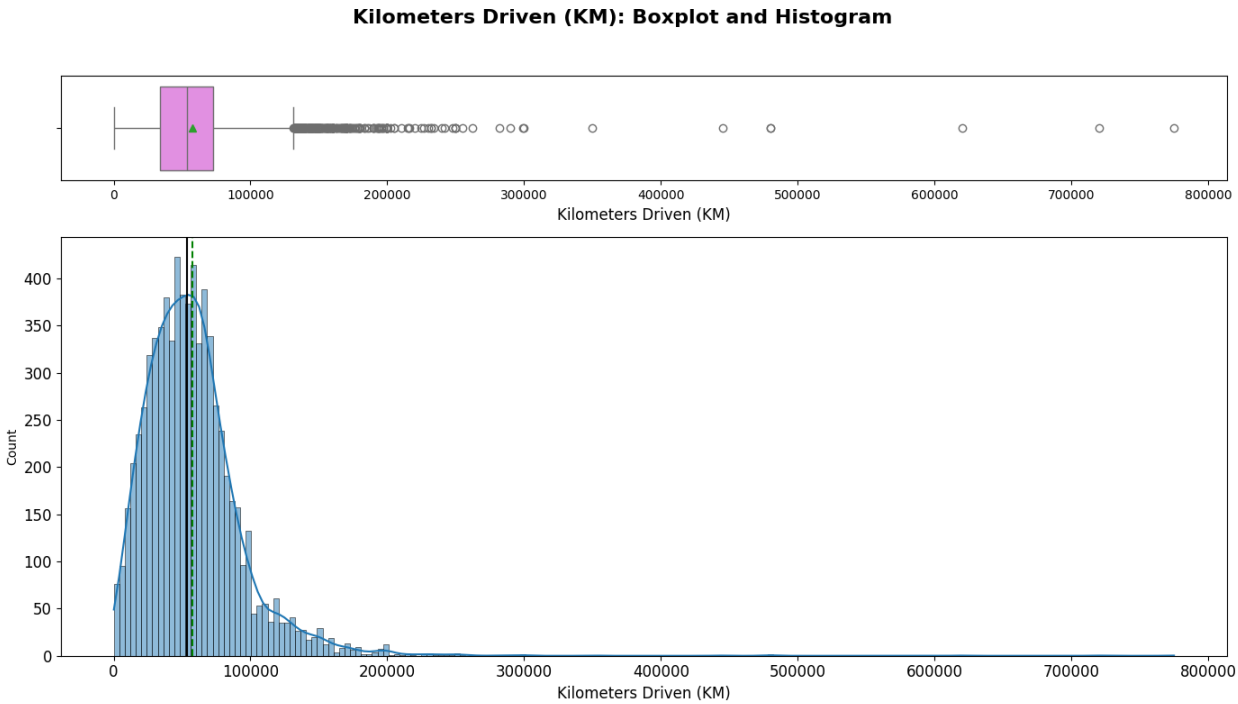
Figure 5. Price (INR 100,000): Boxplot and Histogram



Observations and Insights on Price (INR 100,000): Boxplot and Histogram

Because this variable is the target, preliminary data visualizations are shown before the preprocessing methods are applied. The boxplot and histogram of the Price target variable are heavily right-skewed and contain many outliers. Most used cars are priced lower, ranging from 0.44 to 5.64 INR 100,000, while a few are very expensive, reaching up to 160.00 INR 100,000. The median (denoted by a solid black line in the histogram) is at 9.48 INR 100,000, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 9.48 INR 100,000. The high-priced used cars valued as outliers include luxury-brand models (e.g., BMW, Mercedes, and Audi), newer used cars, high-trim SUVs, and premium diesel models. The histogram suggests that Cars4U's used car inventory primarily consists of budget and mid-range models. The mean is substantially higher than the median, suggesting that luxury and premium used-car models drive up the average price. The long right tail in the Price histogram indicates luxury or high-spec used cars that notably raise the average. Due to its skewness, a log transformation is needed to stabilize variance and reduce the influence of extreme values when building models.

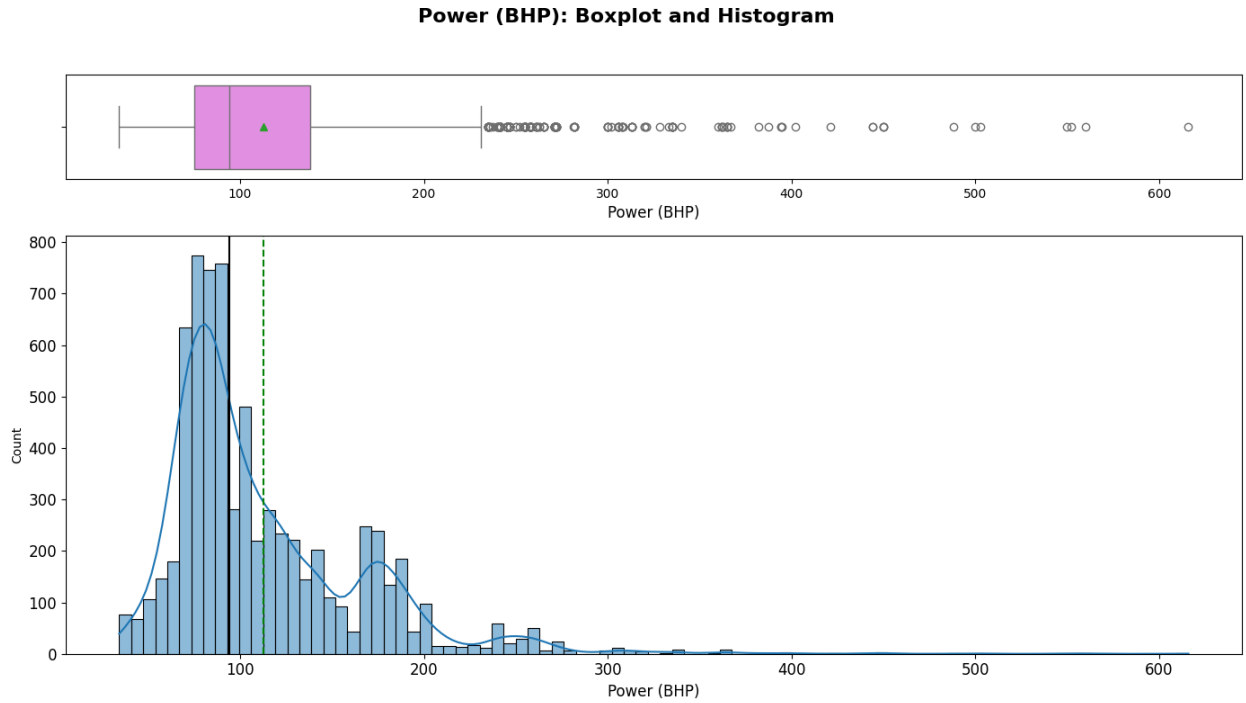
Figure 6. Kilometers Driven (KM): Boxplot and Histogram



Observations and Insights on Kilometers Driven (KM): Boxplot and Histogram

The histogram and boxplot of the Kilometers Driven numerical variable are heavily right-skewed. The majority of used cars driven by previous owners have relatively low to moderate mileage, below 100,000 kilometers, as shown in the boxplot. There are many used cars with extremely high mileage, above 200,000 kilometers, and some even have driven close to 700,000 to 800,000 kilometers. This suggests that these outlier values represent used commercial vehicles, fleet cars, long-distance highway use, and Diesel SUVs with longer engine longevity. The median (denoted by a solid black line in the histogram) is at 53,404 Kilometers Driven, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is at 57,810.85 Kilometers Driven. The histogram indicates that the most commonly used cars have fewer than 100,000 kilometers on the odometer. This is common, as most customers prefer to purchase used cars with fewer than 100,000 kilometers on the odometer. The used cars in the right tail of the distribution are significantly cheaper because they are heavily used, most likely commercial or fleet models, or older models. Due to the existence of extreme values, a log transformation is necessary for modeling.

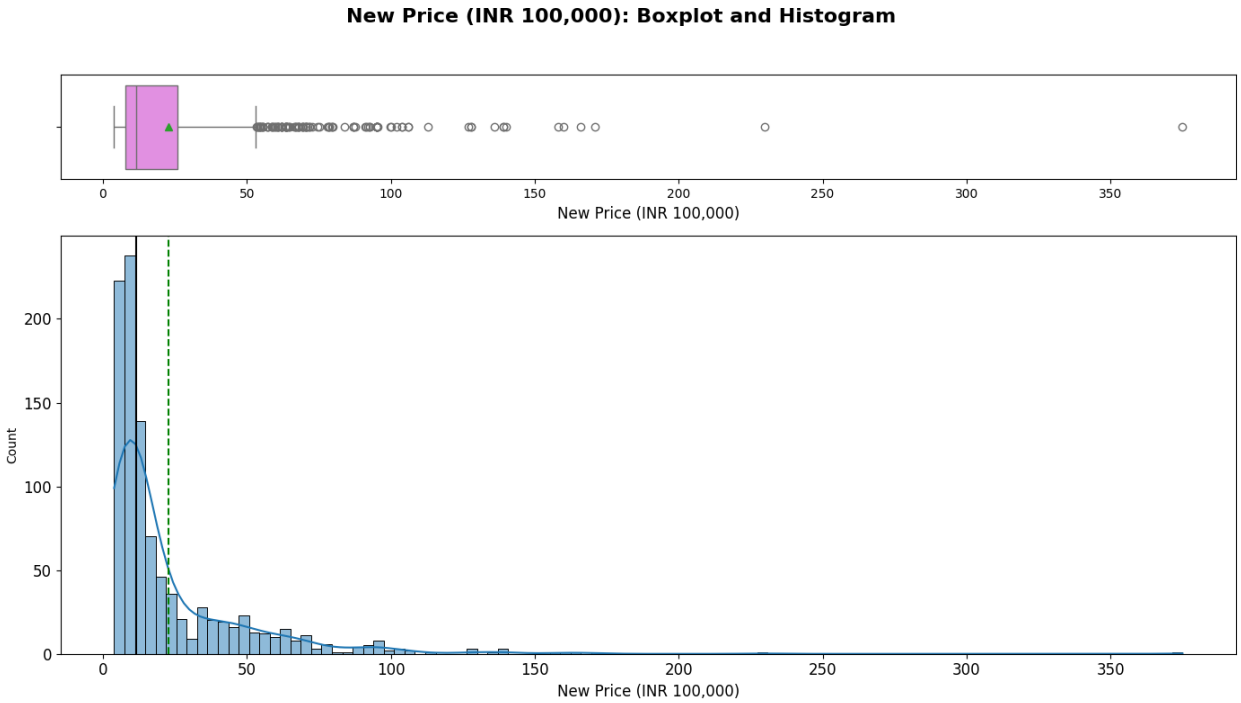
Figure 7. Power (BHP): Boxplot and Histogram



Observations and Insights on Power (BHP): Boxplot and Histogram

The boxplot and histogram for the Power numerical variable indicate a heavily right-skewed distribution. There is a large concentration of used cars with power falling between 70 and 110 BHP. There is also a long right tail of used cars with power values from 150 to 600 BHP. The right-skewed distribution is due to most Indian cars being small hatchbacks and sedans with moderate power. Again, there is evidence of a few high-performance or luxury used cars with higher power. The median (denoted by a solid black line in the histogram) is 94 BHP, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 112.77 BHP. These central tendency measures align with the Indian used-car market, since used cars usually have 1.2L petrol-powered engines producing 80 to 90 BHP and 1.5L diesel-powered engines producing 100 to 115 BHP, rather than larger, more powerful engines.

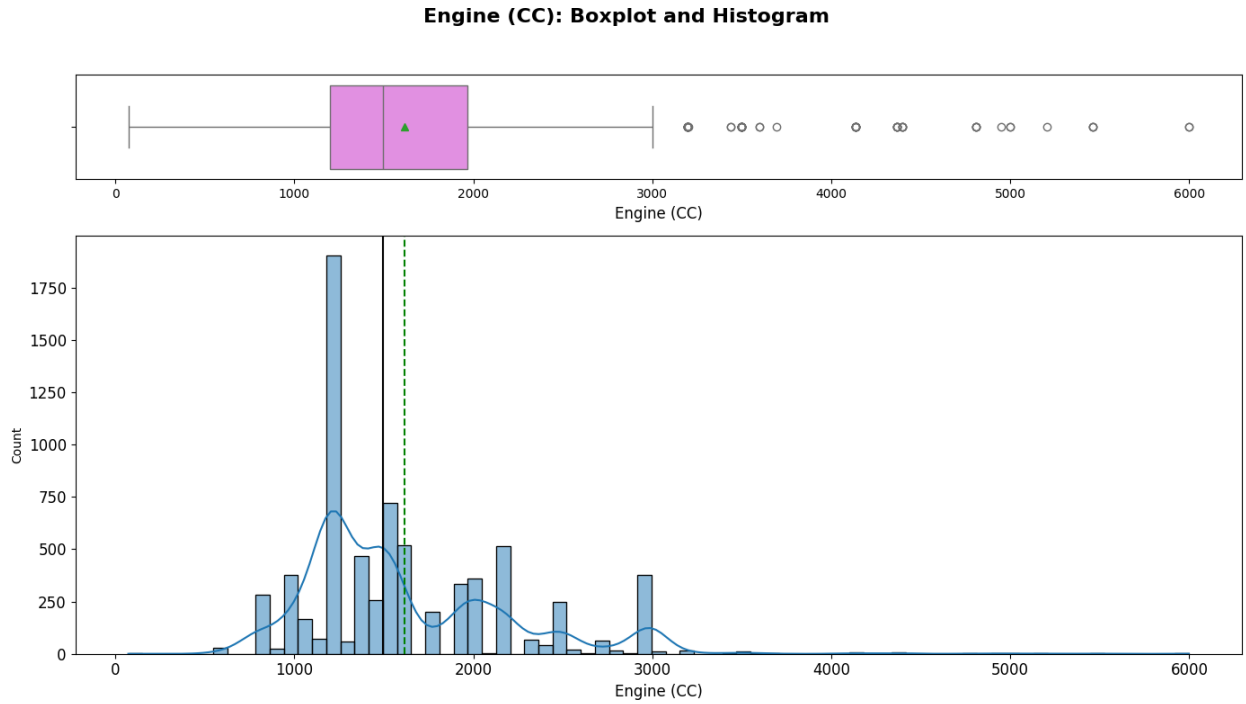
Figure 8. New Price (INR 100,000): Boxplot and Histogram



Observations and Insights on New Price (INR 100,000): Boxplot and Histogram

The New price boxplot and histogram are heavily right-skewed. The median (denoted by a solid black line in the histogram) is 11.57 INR 100,000, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 22.78 INR 100,000. There is a large majority of used cars with new price values below 50 INR 100,000, and a long right tail extending from 100 to 300 INR 100,000. The majority of used cars are low-budget or mid-range. There is further evidence that a small number of used cars are luxury or premium-value, and are considered outliers. Again, these are car brands, including BMW, Mercedes, Audi, Jaguar, High-end SUVs (e.g., Land Cruiser, GLS, and Q7), and luxury sports cars (e.g., Porsche and Mustang). Some important insights include that the initial dataset comprises a wide range of used-car types, high-end used cars disproportionately affect averages, and outliers may carry important business information.

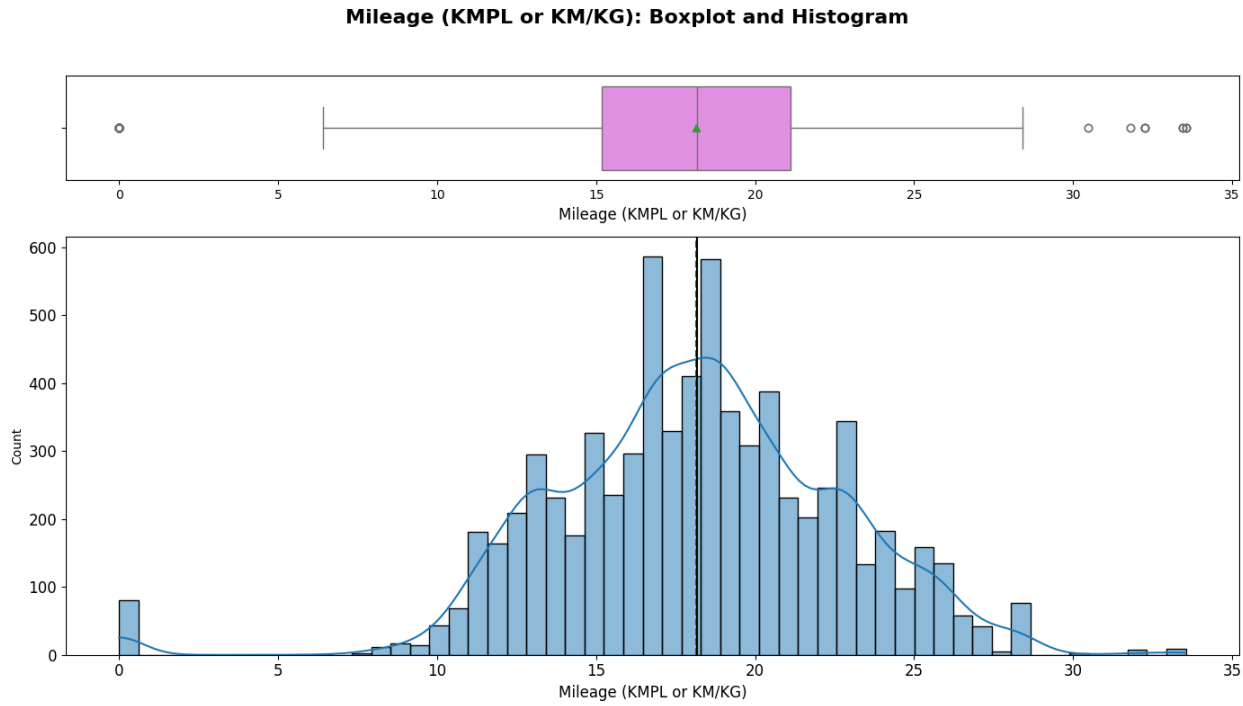
Figure 9. Engine (CC): Boxplot and Histogram



Observations and Insights on Engine (CC): Boxplot and Histogram

The boxplot and histogram of the Engine numerical variable indicate a right-skewed distribution. The core distribution of the Engine numerical variable is stable, centered around plausible values, and unimodal. Most used cars have engine sizes in the 1,000-1,600 CC range. There is a long right tail with higher engine capacities ranging from 2,000 to 6,000 CC, indicating that these used cars are likely typical Indian passenger cars. The majority of mass-market used cars have small- to mid-size engines, whereas SUVs, performance cars, and luxury vehicles are outliers. The median (denoted by a solid black line in the histogram) is 1,493 CC, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 1,616.38 CC. The main mass of the data is also tight and consistent, with two mild peaks around 1,000 and 2,000 CC. There are also high-end outliers beyond 3,000 CC, with some reaching 6,000 CC, most likely luxury performance vehicles, diesel trucks, and specialty models. These central tendency measures align with the Indian used-car market, where hatchbacks and compact sedans dominate, while mid-size engines (1,500-2,000 CC) are less common, and large engines are rare.

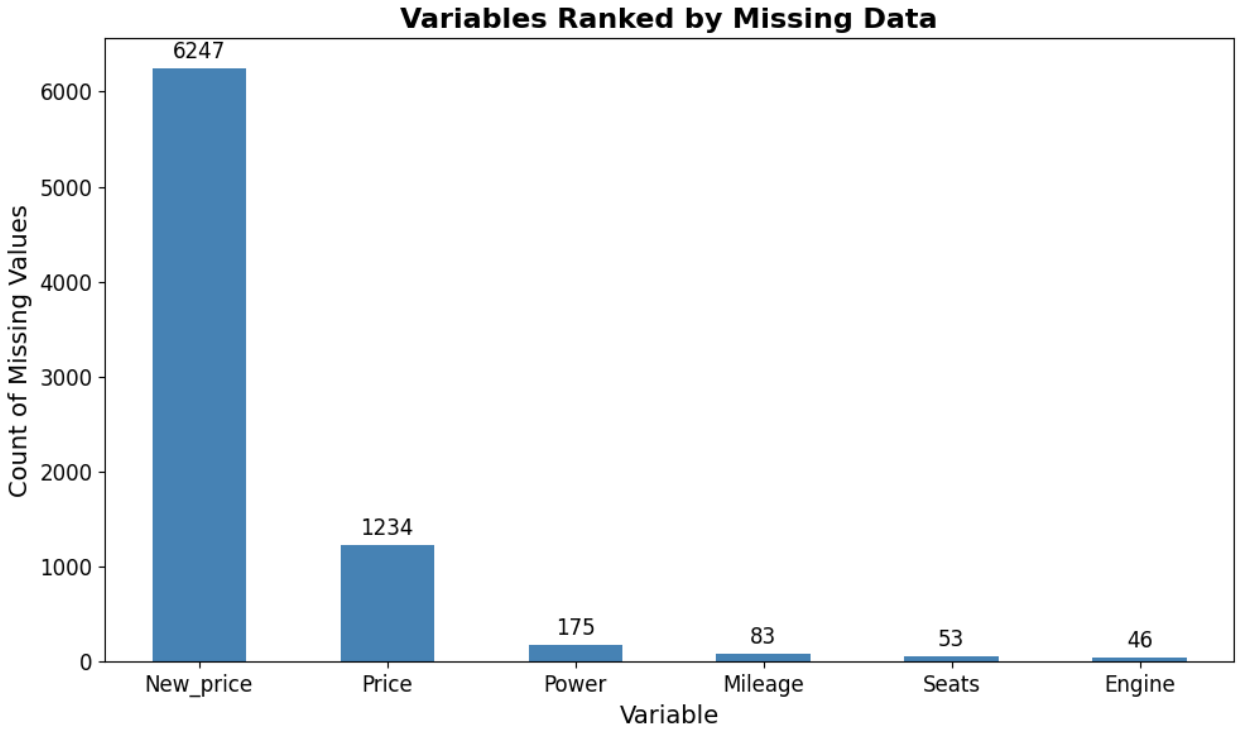
Figure 10. Mileage (KMPL or KM/KG): Boxplot and Histogram



Observations and Insights on Mileage (KMPL or KM/KG): Boxplot and Histogram

The boxplot and histogram for the Mileage continuous numerical variable show that the most commonly used cars have mileage between 15 and 22 KMPL (KM/KG). The distribution is roughly unimodal. There is a long right tail, indicating very high-end outliers. These outliers are a few extremely high-mileage used cars, which could be CNG or hybrid used cars with higher efficiency, or cars with mileage listed in KM/KG rather than KMPL. The median (denoted by a solid black line in the histogram) is 18.16 KM/KG or KMPL, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 18.14 KM/KG or KMPL. The median and mean both fall within the expected range for Indian used cars, suggesting a dominance of petrol and diesel cars. However, the core distribution for the mileage variable is stable and concentrated around realistic values. Some insights include that consumer vehicle efficiency is predictable and that there is limited extreme variation across used-car vehicles.

Figure 11. Variables Ranked by Missing Data Bar chart



Observations and Insights on Variables Ranked by Missing Data Bar Chart

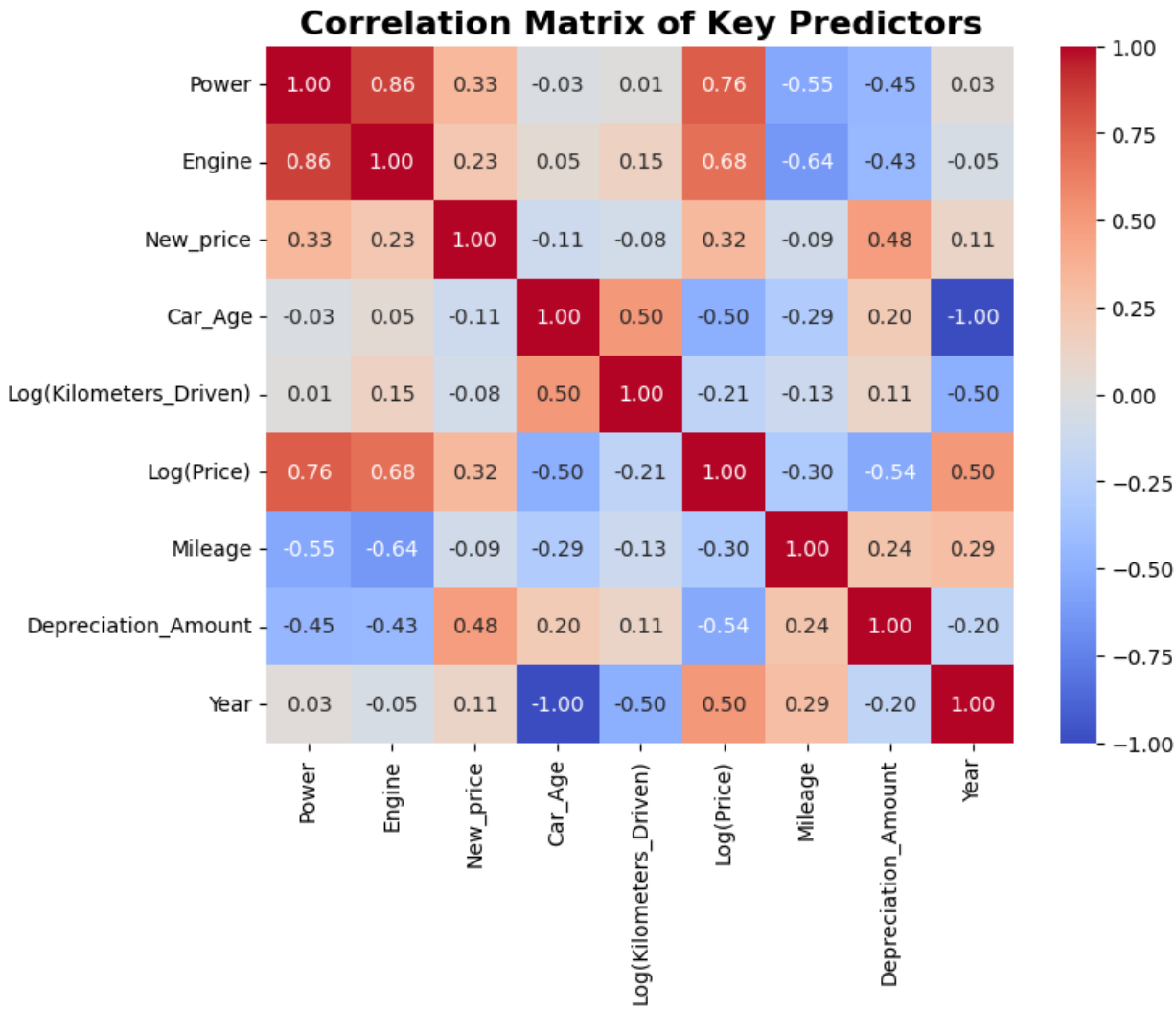
The New price variable has the most missing values of any variable. The target-dependent Price variable also has a significant number of missing values, but far fewer than those in New Price. Mileage, Engine, Power, and Seats have substantially fewer missing values, but are still fewer than New Price and Price. The New price variable is considered a weaker independent variable due to a vast amount of missing values. This may be due to new car prices not being recorded for older models. The missing values in the Price variable must be removed, reducing the usable dataset size for modeling. Mileage, Engine, Power, and Seats need minimal data cleaning, which is preferable for model stability. The missing values for Mileage are now updated to include data entry error values of "0.0." The clustered pattern of missing values in certain columns indicates non-random missingness that may be tied to specific car brands or models, older vehicles, or incomplete listings.

Third, strong, interpretable relationships were identified during the exploratory data analysis of the dataset. The bivariate and correlation analyses identified multiple consistent, intuitive pricing-independent variables (predictors). It is important to note that the target Price variable was log-transformed to reduce skewness, improve model stability, and ensure homoscedasticity and greater linearity in relationships with other

independent variables (predictors), thereby making the models more suitable for regression. It is also important to note that the Kilometers Driven independent variable was also log-transformed to reduce or eliminate its skewness. Other reasons for this log transformation include that Kilometers Driven spans several orders of magnitude and has a non-linear, diminishing-returns relationship with the target price variable. There was a significant positive correlation between New Price and Price log, with a Pearson correlation coefficient of 0.32. Also, there was a significant positive correlation between Power and Log(Price), with a Pearson correlation coefficient of 0.76. Moreover, there was a significant positive correlation between Engine and the Log(Price), with a Pearson correlation coefficient of 0.68. There was also a negative correlation between Log(Kilometers Driven) and Log(Price), with a Pearson correlation coefficient of -0.21. Also, there was a negative correlation between Mileage and the Price(Log), with a Pearson correlation coefficient of -0.30. Furthermore, there was a negative correlation between Car Age and the Log(Price), with a Pearson correlation coefficient of -0.5. Overall, a key finding from the Exploratory Data Analysis is that the strongest independent variables (i.e., predictors) of price (e.g., Engine, Power, Car Age, and New Price) align with real-world automotive economics.

The data visualizations below depict the significant, interpretable relationships in the correlation structure and scatterplots of the Log(Price) dependent variable with the important independent variables. The first correlation heatmap of the top predictors visually shows the economic logic behind pricing. The second through eighth visualizations depict the key independent variables (features) that have strong bivariate economic relationships with the target variable, Log(Price). When analyzed together, variables tied to a car's intrinsic value (e.g., Power, New Price, Engine, and Year) indicate clear positive relationships with Price log. In contrast, variables related to wear (e.g., Car Age, Kilometers Driven, and Mileage) show negative relationships with Log(Price). Altogether, these scatterplots suggest a coherent pricing structure: newer, more powerful, and originally expensive used cars are priced higher. In contrast, heavily driven vehicles are expected to depreciate.

Figure 12. Correlation Heatmap of Key Predictors Heatmap

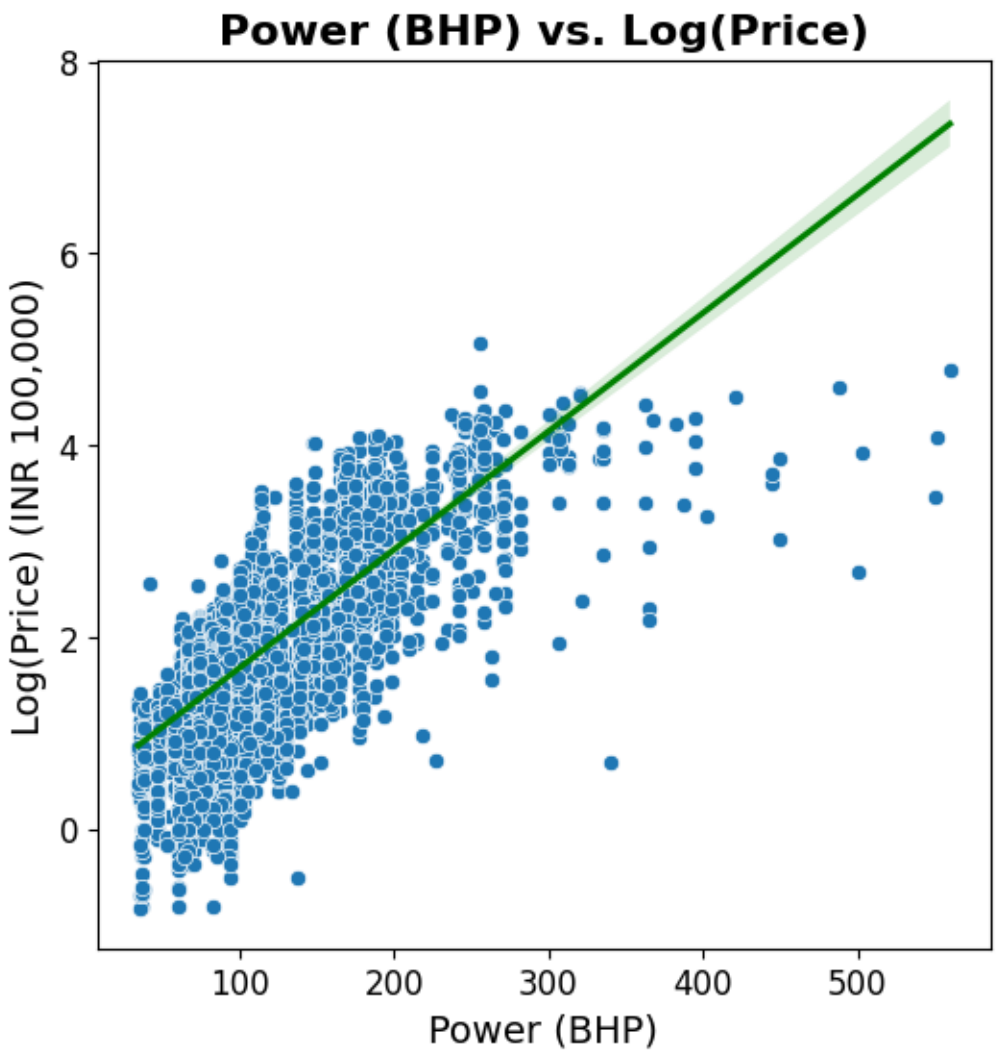


Observations and Insights on Correlation Matrix of Key Predictors Heatmap

Depreciation Amount aligns robustly with Car Age, Mileage, and Kilometers Driven log. To clarify, Car Age and Depreciation Amount have a positive Pearson correlation coefficient of 0.20, indicating that older used cars tend to depreciate more. Mileage and Depreciation Amount have a positive Pearson correlation coefficient of 0.24, indicating that higher mileage is associated with higher depreciation. The Log(Kilometers Driven) and Depreciation Amount have a positive Pearson correlation coefficient of 0.11, reinforcing the aforementioned patterns. This confirms that depreciation is motivated by usage and age. Power and Engine both correlate strongly with Log(Price), with Pearson correlation coefficients of 0.76 and 0.68, respectively. Instead, Power and Engine have weaker correlations with Depreciation, at -0.45 and -0.43, respectively. This means that high-spec used cars start expensive and remain so as they depreciate, but not disproportionately so. The New price variable is

characterized as a bridge variable, showing a positive correlation with Log(Price) (Pearson correlation coefficient of 0.32). The New price variable also has a positive correlation with Depreciation Amount, with a Pearson correlation coefficient of 0.48. These correlations indicate that a higher original manufacturer suggested retail price still affects resale value, and expensive used cars can decline in dollar value, even if they still maintain value better in percentage terms. Mileage has negative correlations with Power and Engine, with Pearson correlation coefficients of -0.55 and -0.64, respectively. This indicates that Mileage is inversely related to performance, particularly for higher-spec used cars, which tend to be driven much less or have better maintenance. Car Age and Mileage are moderately correlated with a Pearson correlation coefficient of 0.50. Year has positive correlations with Log(Price), Power, New price, and Mileage, with Pearson correlation coefficients of 0.50, 0.03, 0.11, and 0.29, respectively. This implies that newer cars are usually more powerful, have larger engines, have higher original prices, and therefore have higher resale prices. Year has a negative correlation with Engine and Log(Kilometers Driven), with Pearson correlation coefficients of -0.05 and -0.50, respectively. This implies that newer used cars have had less time to accrue driven kilometers and are therefore more highly used. Lastly, Year has a negative correlation relationship with Depreciation Amount, with a Pearson correlation coefficient of -0.20. This implies that depreciation is cumulative: relatively new used cars have depreciated by much less, while older used cars have depreciated over time.

Figure 13. Power (BHP) vs. Log(Price) Scatterplot

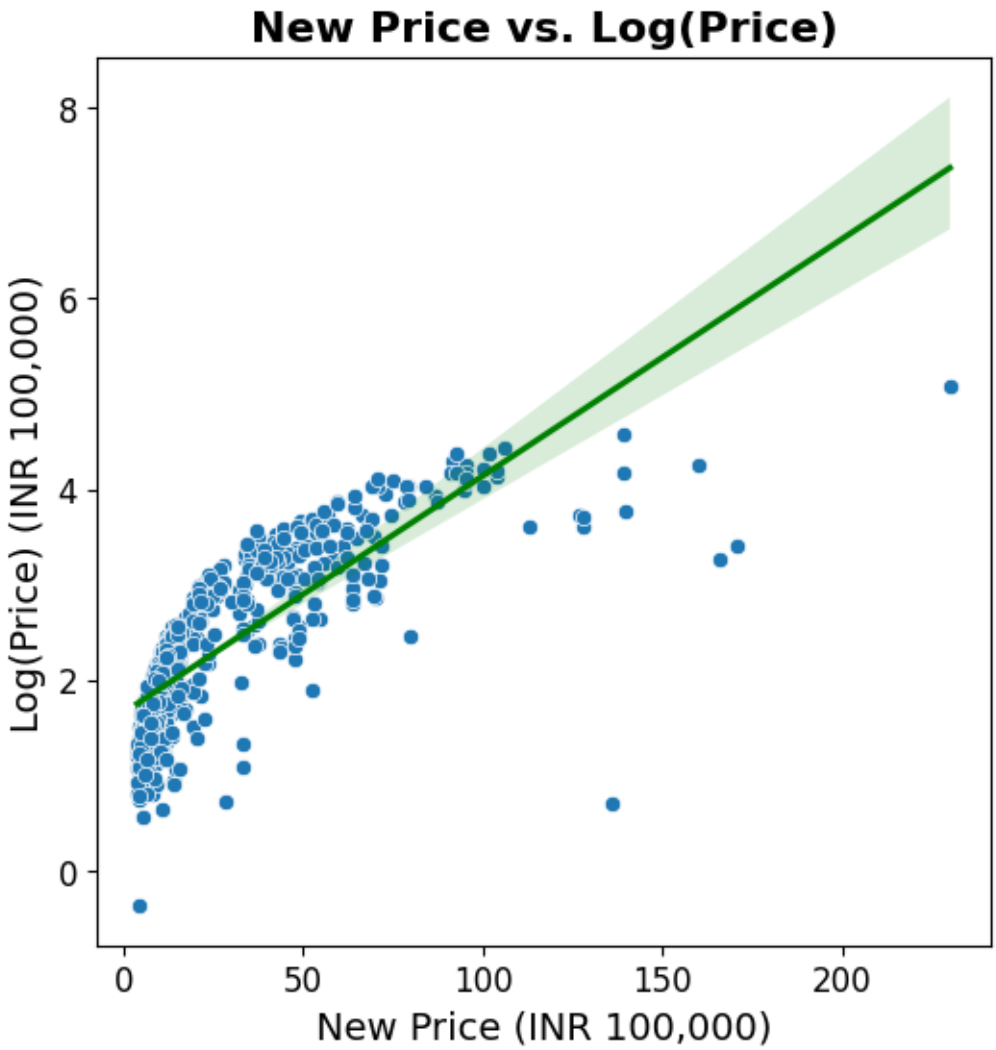


Observations and Insights on Power vs. Log(Price) Scatterplot

The Power vs. Log(Price) scatterplot indicates a strong positive relationship (Pearson correlation coefficient of 0.76) between used-car power and Log(Price). As a used car's horsepower increases, its log price increases. Hence, the regression line clearly captures this pattern. Even though the overall relationship is linear, there is evidence of vertical variability that widens at higher power levels, which realistically reflects the diversity of higher-performance and luxury used-car models that command different price premiums even at comparable power outputs. Otherwise, these used cars have more powerful engines and command higher log prices. Even though this vertical variability exists, the basic relationship holds: power is a significant and consistent variable that effectively determines the price of a used car. Most importantly, there is no evidence of clustering or segmentation in the scatterplot,

suggesting that power influences price across the entire dataset rather than only within specific clusters or subgroups.

Figure 14. New Price vs. Log(Price) Scatterplot

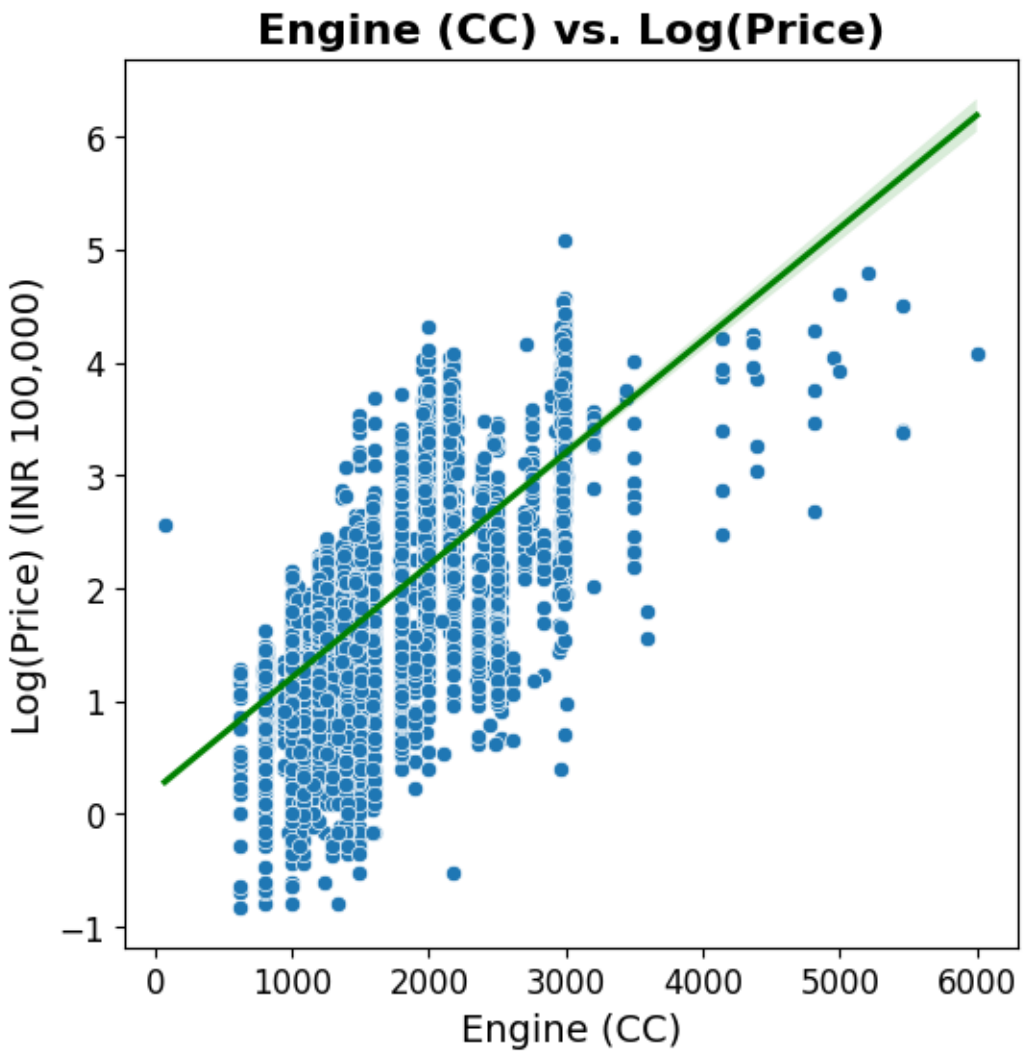


Observations and Insights on New Price vs. Log(Price) Scatterplot

The New Price vs. Log(Price) scatterplot indicates a strong, smooth positive relationship (Pearson correlation coefficient of 0.32) between a used car's original new car price value and its current log-transformed price value. While the new price increases, the Log(Price) values also increase. The regression line also fits the data points well. In other words, used cars that were initially costly tend to have higher resale values. This also aligns with the fact that these specific used cars, which were

once expensive, still command higher prices despite depreciation and market variability. There is further evidence that the vertical spread widens at higher new price values, indicating the presence of premium and luxury used cars in the collected dataset. As such, these premium and luxury used cars have more varied resale values, influenced by factors such as brand reputation, model year, mileage, and condition. Overall, New price is one of the most potent predictors of used-car prices in the dataset, and the Log(Price) helps visualize this relationship more clearly.

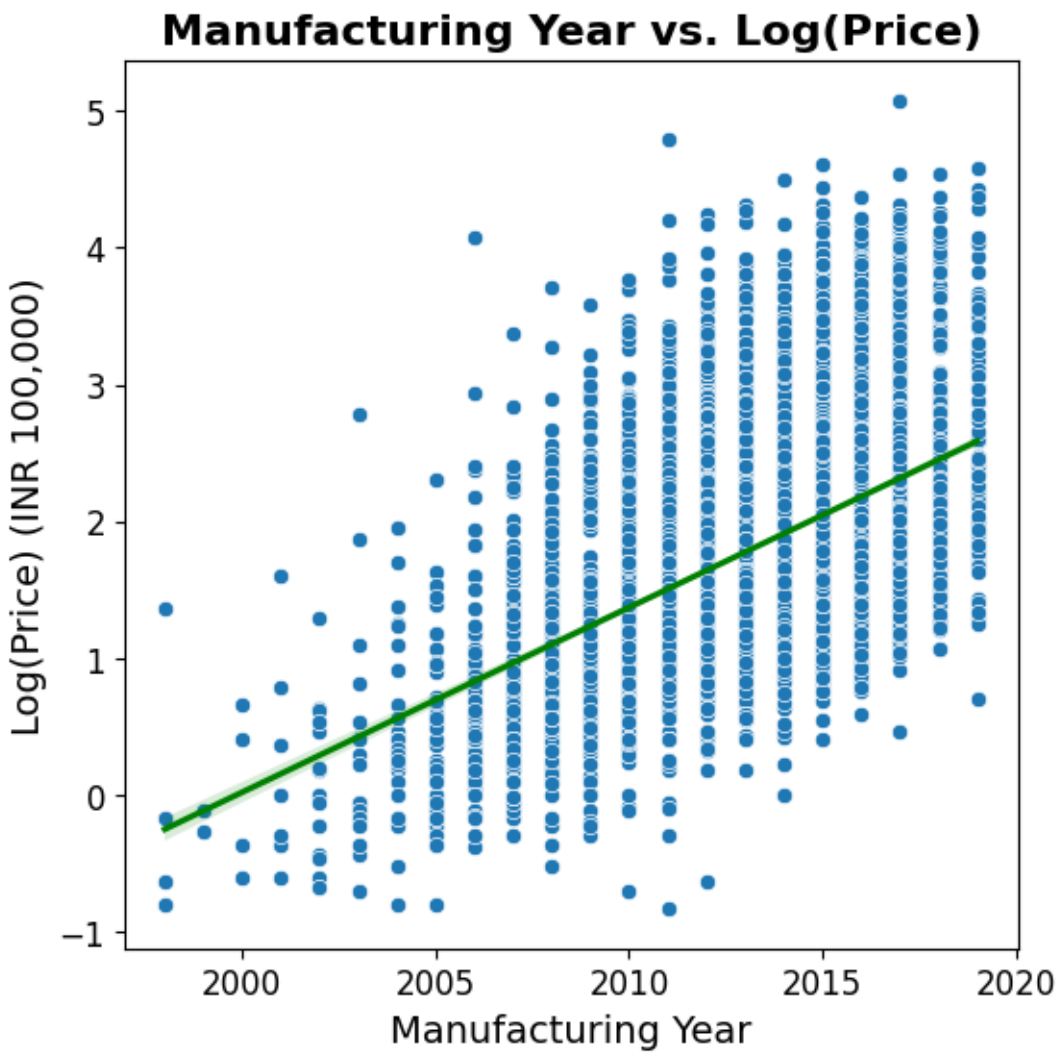
Figure 15. Engine (CC) vs. Log(Price) Scatterplot



Observations and Insights on Engine (CC) vs. Log(Price) Scatterplot

The Engine vs. Log(Price) scatterplot indicates a clear positive relationship (Pearson correlation coefficient of 0.68) between engine size and the Log(Price) of used cars. Used cars with larger engine displacements tend to be more expensive. The regression line indicates an upward trend, suggesting that engine displacement can be a significant predictor of the target price value even after applying a log transformation. Even though the overall pattern is positively linear, there is evidence of a vertical spread that widens for used vehicles with larger engines, mirroring market dispersion. Specifically, premium used-car brands, performance trims, and luxury segments cluster in this region, affecting price variability. Although there is evidence of dispersion, engine size remains a significant predictor of used-car price. From 800 to 2,500 CC, there are strong, tightly clustered points around the line of best fit, and from beyond 3,000 CC, the spread widens significantly. More importantly, higher-capacity used cars with more powerful engines (e.g., 1800 CC, 2000 CC, and 2400 CC) correspond to higher Log(Price) values. Thus, the relationship is smooth and expected across the range of values along the regression line.

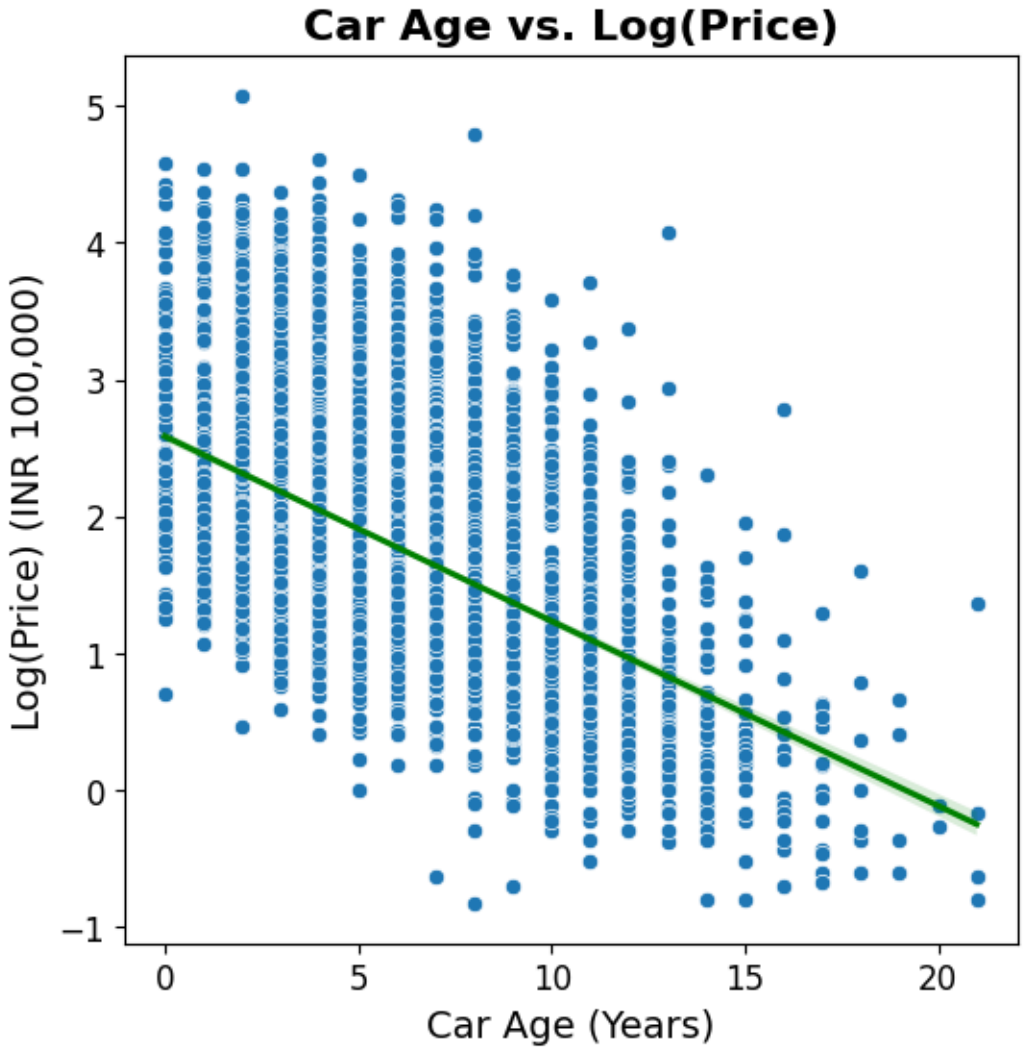
Figure 16. Manufacturing Year vs. Log(Price) Scatterplot



Observations and Insights on Manufacturing Year vs. Log(Price) Scatterplot

The Manufacturing Year vs. Log(Price) scatterplot shows a clear positive correlation between the manufacturing year and Log(Price) of a used car. The regression line represents a percentage change in price per additional year, rather than a fixed amount of INR 100,000. There is notable clustering of used cars from 2014 to 2019 and greater variability from 1998 to 2008. In simpler terms, the variances increase as used cars get older. The relationship is robust, but not absolutely linear. Overall, Log(Price) increases with the manufacturing year, suggesting that newer used cars have higher price values in INR 100,000 terms even after a log transformation. The relationship is clearly linear, even though older used cars exhibit greater variability; this is most likely due to differences in condition and past usage.

Figure 17. Car Age vs. Log(Price) Scatterplot

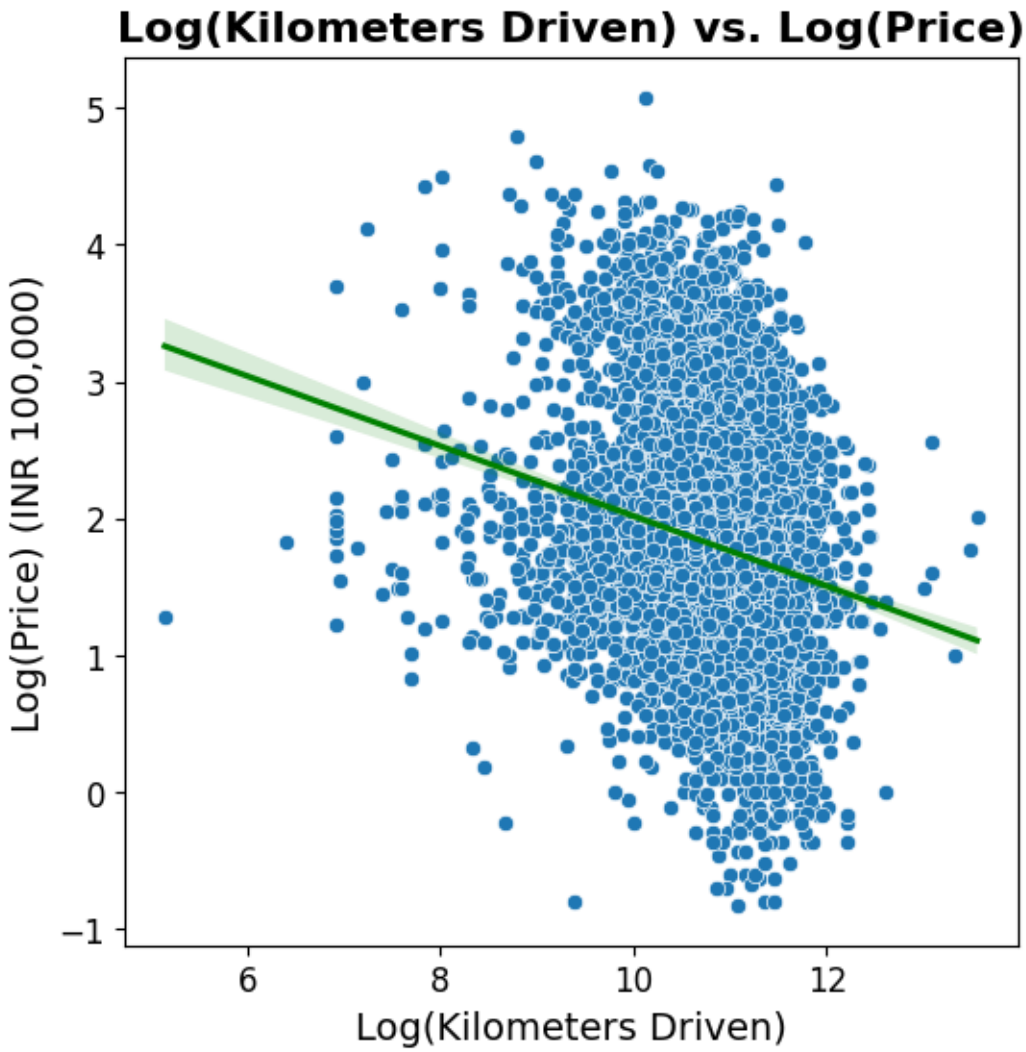


Observations and Insights on Car Age vs. Log(Price) Scatterplot

The Car Age vs. Log(Price) scatterplot shows a clear negative linear relationship (Pearson correlation coefficient of -0.5), with the price of a used car steadily declining as it ages. There is evidence that older cars sell for lower prices, even after applying a log transformation to the target variable. As for younger used cars, most points are clustered closer to the regression line, indicating a rate of early-life depreciation. Older cars show a greater price spread, likely due to differences in condition and to uncommon, infrequent models. In fact, the downward trend line validates that age is one of the most crucial variables in used-car pricing. Specifically, there is evidence that depreciation is rapid in the first few years and slows as used cars age. Additionally,

there is evidence that older cars exhibit greater pricing ambivalence, with differences in condition, brand or model durability, maintenance history, and mileage variation important factors influencing greater price dispersion. Overall, Car Age is a more robust predictor of used-car prices, with older vehicles introducing more noise into the model.

Figure 18. Log(Kilometers Driven) vs. Log(Price) Scatterplot

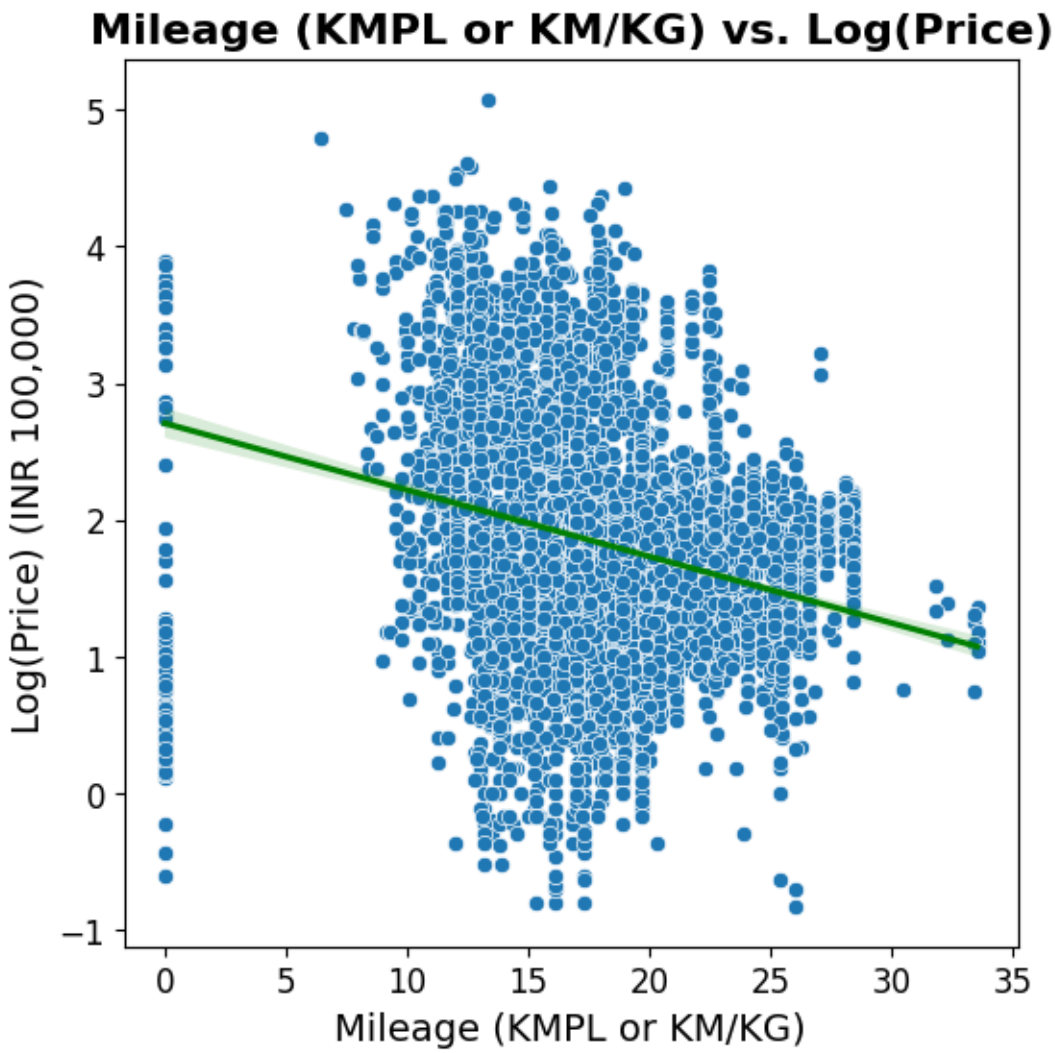


Observations and Insights on Log(Kilometers Driven) vs. Log(Price) Scatterplot

The Log(Kilometers Driven) vs. Log(Price) scatterplot shows a clear negative correlation (Pearson correlation coefficient of -0.21) between Log(Kilometers Driven) and Log(Price) for used cars. As the number of kilometers driven increases, the price of

a used car decreases. The log-transformed data for the two variables show a linear trend. The regression line fits the points closely, suggesting that the total number of kilometers driven by a previous owner is a significant independent variable in a used car's price. There is also evidence of vertical spread at each level of kilometers driven, implying key differences in brand, age, condition, and engine features, but the overall pattern remains stable. Overall, the visualization confirms that higher usage of a used car is associated with lower used-car values, and that the log transformation effectively preserves variance, making the relationship suitable for subsequent linear modeling.

Figure 19. Mileage (KMPL or KM/KG) vs. Log(Price) Scatterplot



Observations and Insights on Mileage (KMPL or KM/KG) vs. Log(Price) Scatterplot

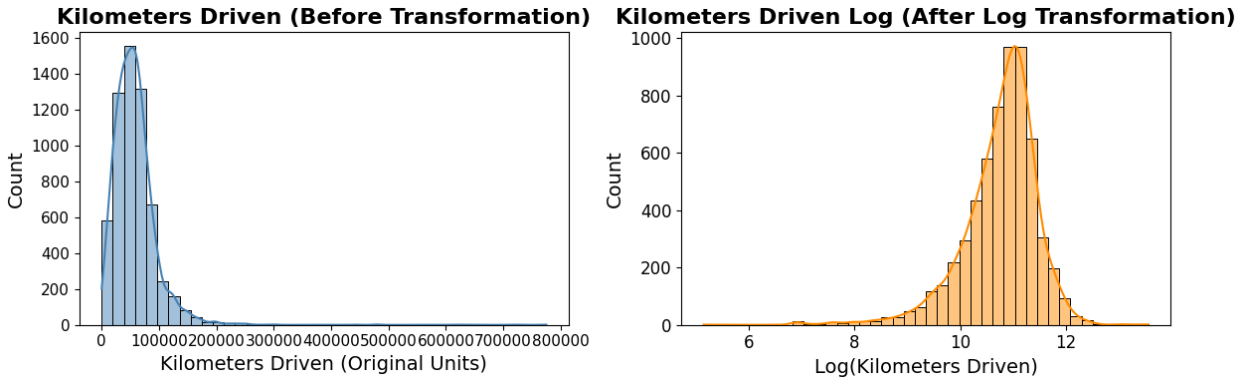
The Mileage vs. Log(Price) scatterplot indicates a clear negative relationship between mileage and the log-transformed price variable, with a Pearson correlation coefficient of -0.30. As used cars accrue more mileage, their value declines in a predictable, consistent pattern. In particular, higher-mileage cars are typically cheaper, with a steeper decrease in log price at lower mileage and a flatter downturn at higher mileage. The regression line clearly represents this downward trend. The log transformation helps stabilize the variance, suggesting a slight linear depreciation effect. Even though there is notable variability in vertical spread at each mileage level, this variability reflects differences in brand, age, condition, and engine across used cars. Thus, the scatterplot confirms that mileage can also effectively predict the price of a used car, as it clearly indicates long-term vehicle wear and lower prices. Some insights include that high-mileage used-cars exhibit lower price variance, whereas low-mileage used-cars exhibit wide price dispersion.

Fourth, the conducted feature engineering techniques further reinforced the dataset's quality. Foremost, it is essential to note that the log transformations applied to the Kilometers Driven independent variable and the target Price dependent variable are crucial feature engineering techniques that make the distributions of both variables more normal and linear, thereby increasing the predictive power of the proposed variables. The feature-engineered independent variables, including Brand, Car Age, and Depreciation Amount, added supplemental, meaningful structure. To clarify, Car Age is a much more interpretable transformation of Year (i.e., the car's manufacturing year). Additionally, the depreciation amount better summarizes several independent variables, such as brand strength, model quality, age, and mileage. In general, key findings from feature engineering techniques include enhanced interpretability and model preparedness through reduced skewness and the capture of economic behavior.

The data visualizations below depict the significant feature engineering techniques and their respective impact. First, the histograms of Kilometers driven before and after log transformation show that the variable becomes less skewed, more linear, and more predictive. Similarly, the second set of Price histograms, before and after the log transformation, also highlights the need to make the distribution more normal and less dominated by outliers, so that the proposed models can better learn

the target variable. The third Car Age boxplot and histogram visualizations show that the dataset is dominated by relatively new cars, with older used cars as outliers, and that older cars depreciate more. The fourth Depreciation Amount vs. Price log-scatterplot shows that older cars depreciate more. The fifth Brand-Level Depreciation Chart shows that specific brands experience varying levels of depreciation, which affect their overall resale value. Together, Car Age and depreciation move in tandem, with older cars losing more value and appearing lower on the Price log axis.

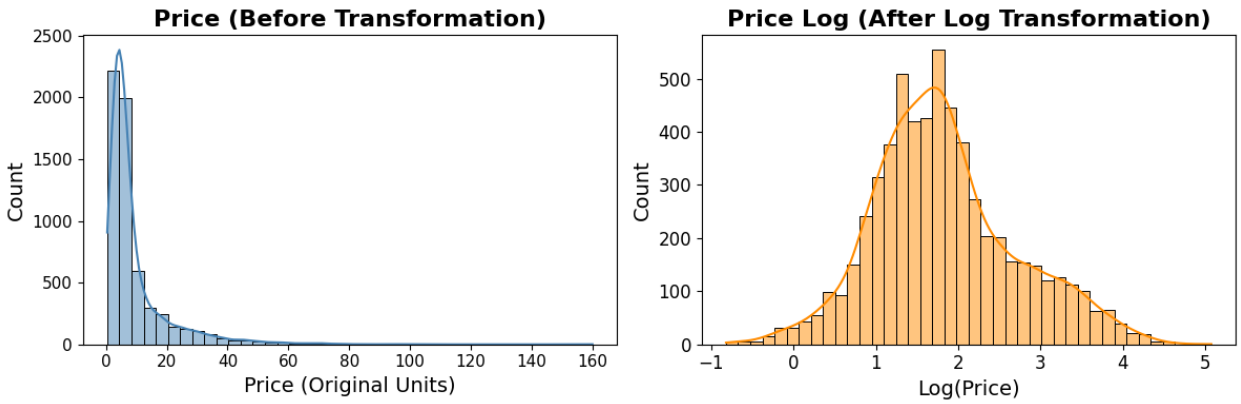
Figure 20. Kilometers Driven Before and After Log Transformation



Observations and Insights on Kilometers Driven Before and After Log Transformation Figure

Prior to a log transformation, Kilometers Driven is significantly right-skewed, with most used-car vehicles having mileage below 100,000 kilometers. There is a long right tail that extends to extremely high values, reaching several hundred thousand kilometers. The scale found in the figure makes it complicated to interpret the central pattern. After the log transformation, the distribution becomes more symmetric, resembling a bell curve. The right tail becomes more compressed, lowering the influence of outliers. The transformed values instead cluster around a secure center, making the variable now readily available for model building. Overall, some essential insights include that the log transformation normalizes the distribution, makes outliers more manageable, linearizes relationships with Price, and enhances interpretability for stakeholders.

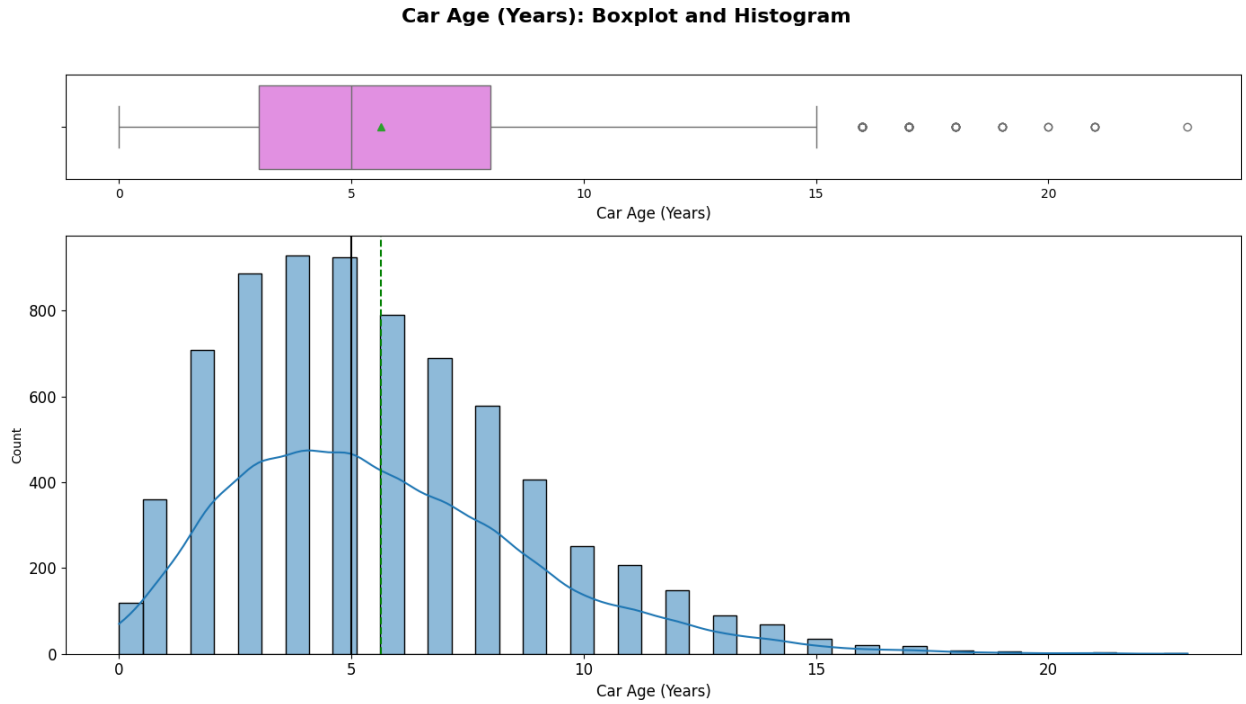
Figure 21. Price Before and After Log Transformation



Observations and Insights on Price Before and After Log Transformation Figure

Prior to a log transformation, Price is heavily right-skewed, with most used-car vehicles priced at the lower end and a long right tail indicating higher-priced vehicles. There is a density curve with a sharp peak near low prices and a slow decline toward higher prices. The scale is wide, leading to interpretability issues for central tendency. After the log transformation, the distribution is symmetric, resembling a bell curve. The right tail is now compressed, reducing the influence of luxury-priced vehicles, and the central mass of the data is clearer and more interpretable. The variable is now suitable for modeling building. Overall, some essential insights include that the log transformation normalizes the distribution, makes outliers more manageable, makes relationships with other independent variables (predictors) more linear, and enhances interpretability for stakeholders.

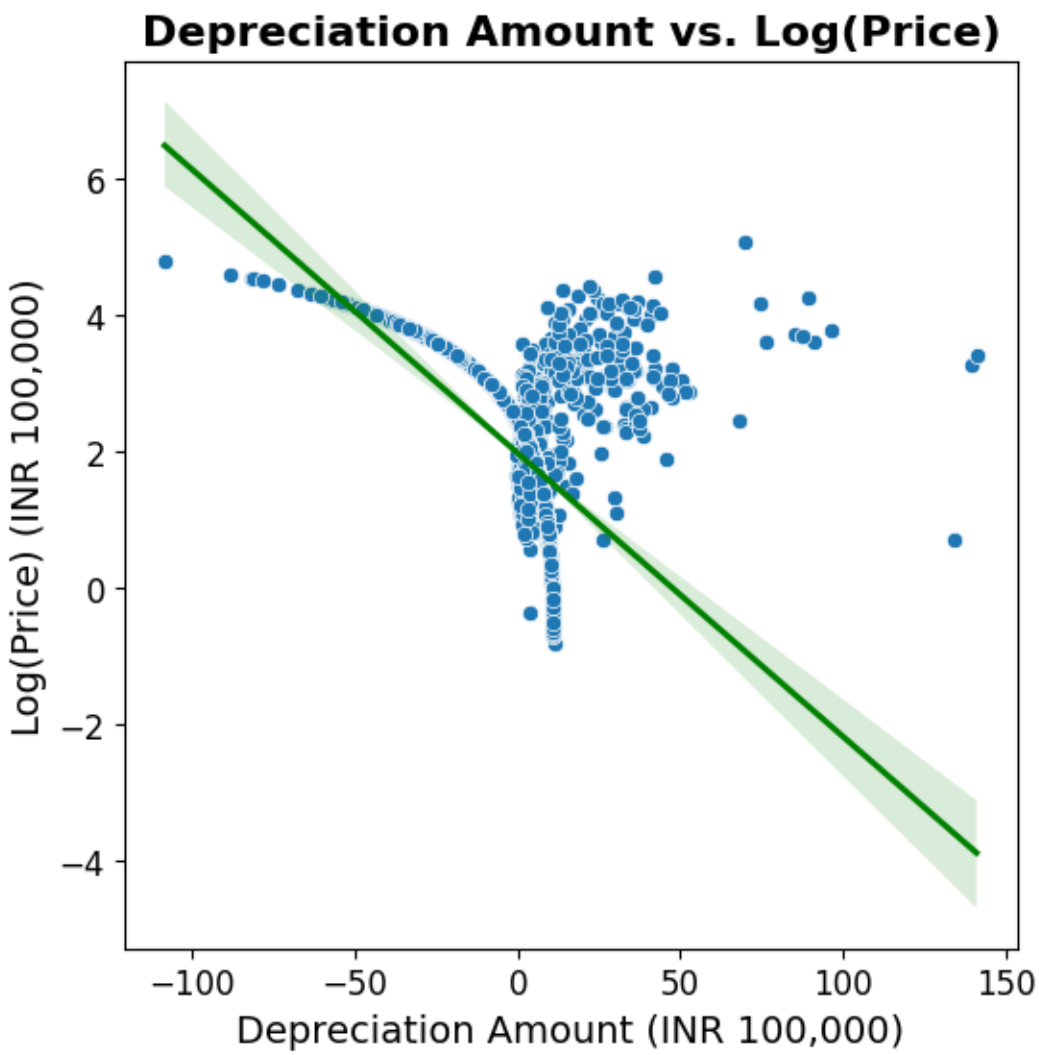
Figure 22. Car Age (Years): Boxplot and Histogram



Observations and Insights on Car Age (Years): Boxplot and Histogram

The boxplot and histogram of the feature-engineered Car Age variable show that the most commonly used cars are between 2 and 8 years old. The distribution is right-skewed. There is a long right tail, indicating a notable number of older used cars, as seen in the outliers. very high-end outlier values. These outliers represent used cars that are mid-teen or older, which are less common. The median (denoted by a solid black line in the histogram) is 5, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 5.64. Overall, Car Age shows a concentrated cluster of young- to mid-age used cars, with a long right tail of older outliers. This right-skewed pattern best captures the dynamics of depreciation and explains variation in used-car prices.

Figure 23. Depreciation Amount vs. Log(Price) Scatterplot

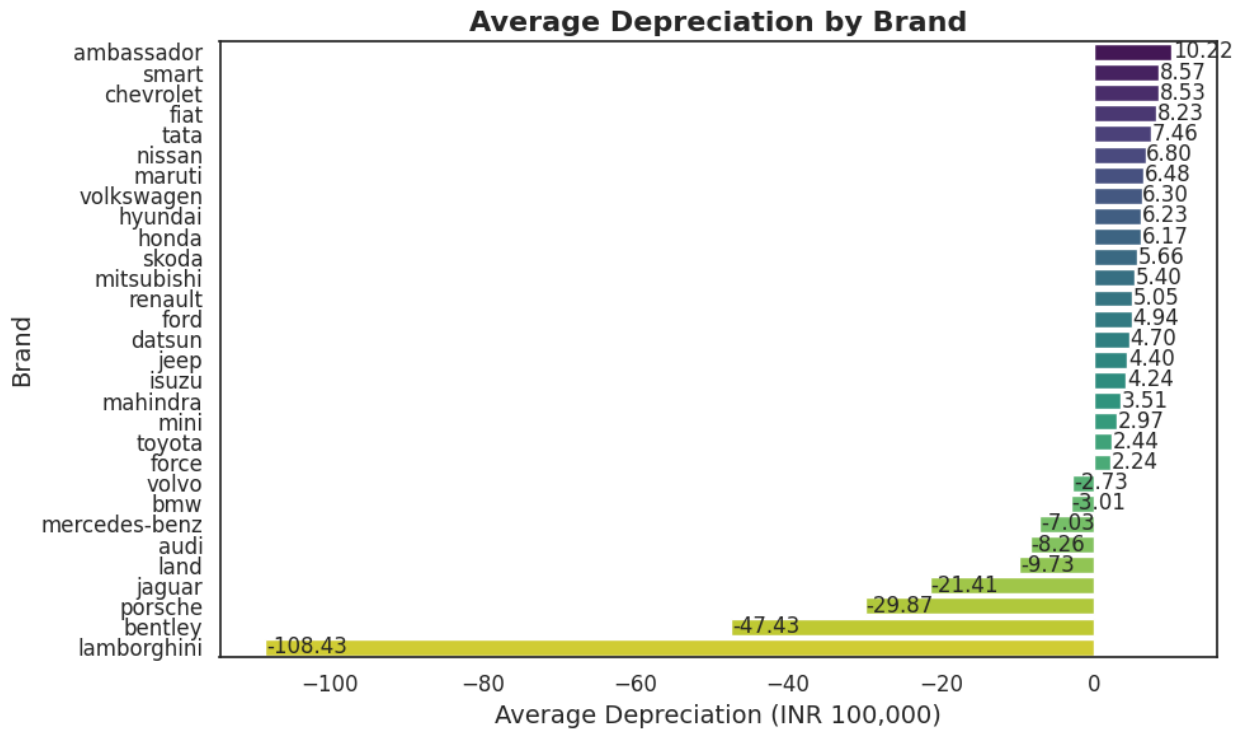


Observations and Insights on Depreciation Amount vs. Log(Price) Scatterplot

The Depreciation Amount vs. Log(Price) scatterplot shows a strong negative linear relationship. As depreciation increases, Log(Price) decreases. The regression line slopes downward, indicating that used cars with greater depreciation usually have lower prices. Using Log(Price) straightens the relationship between depreciation and price, yielding a more linear, model-friendly visualization. There is evidence that low-depreciation used cars cluster at much higher log prices, suggesting newer or value-retaining models, typically associated with strong or premium brands. There is also evidence of high-depreciation vehicles that show greater variability, suggesting realistic variation in how luxury or older used cars depreciate. Overall, the scatterplot supports the conclusion that Depreciation Amount is a robust explanatory variable,

showing that used cars that depreciate more from new to used tend to be priced lower. The log transformation indicates a clear linear relationship.

Figure 24. Average Depreciation by Brand Chart



Observations and Insights on Average Depreciation by Brand Chart

The following key observations and insights can be made from the Brand-Level Average Depreciation chart. First, luxury brands such as Lamborghini, Bentley, Porsche, Jaguar, Land Rover, Audi, BMW, and Mercedes-Benz have the highest absolute depreciation, ranging from -108.43 to -7.03. Luxury vehicles usually have higher maintenance and ownership costs, which accelerate value loss. Second, mass-market brands such as Maruti, Tata, Hyundai, Honda, Nissan, Chevrolet, Fiat, Smart, and Ambassador have much lower depreciation rates and retain their value better. These used cars have average depreciation ranging from 6.48 to 10.22. These brands have stronger value retention, with lower initial purchase prices, strong demand in the used-car market, and lower maintenance costs. Third, mid-tier global brands, such as Volkswagen, Ford, Renault, Skoda, Mitsubishi, Jeep, Mahindra, Toyota, and Volvo, exhibit reasonable depreciation, with predictable, steady patterns. These used-car brands have average depreciation rates ranging from 2.24 to 6.30. These brands also

have more stable demand in the used-car market. Fourth, economy brands have the lowest depreciation, mainstream brands have moderate depreciation, and luxury brands have the highest depreciation. Altogether, brand identity is an important driver of value loss in used cars.

A brief discussion that explicitly explains the different types of regression models is provided before advancing with the general performance overview of each regression model. An Ordinary Least Squares (OLS) Regression (Linear Regression) model fits a straight line to the data to capture a basic trend. It assumes a simple, direct relationship between variables. An L2-regularized linear regression (Ridge Regression) model is a linear model that penalizes large coefficients, helping prevent overfitting to noise in the data. An L1-Regularized Linear Regression (Lasso Regression) model is similar to a Ridge Regression model, but is uniquely capable of entirely ignoring (dropping) less important variables to simplify the model automatically. A Decision Tree (Decision Tree Regression) model uses a flowchart-like structure, repeatedly splitting the data into smaller, distinct groups based on specific criteria to produce a prediction. A Random Forest Regression model (Random Forest) builds hundreds of independent Decision Trees and averages their collective predictions, significantly improving overall stability and accuracy. A k-Nearest Neighbors Regression (KNN Regression) model predicts outcomes by identifying the closest, most similar historical data points (its “neighbors”) and averaging their values.

Another brief discussion of how each regression model’s performance was assessed is provided below. Each regression model was assessed using a consistent performance assessment protocol (a get_model_score function) that compared predicted values with the actual target values on both the training and test datasets. Predictions were produced for each split, converted back to the original Price scale from the Log(Price) transformed scale when the target variable had been log-transformed, and then assessed using standard regression metrics. The performance metrics assessment comprised R-squared and Adjusted R-squared to measure explained variance, as well as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) to measure prediction error. Overall, these assessed performance metrics guaranteed a balanced view of model fit, generalization performance, and error significance for each model.

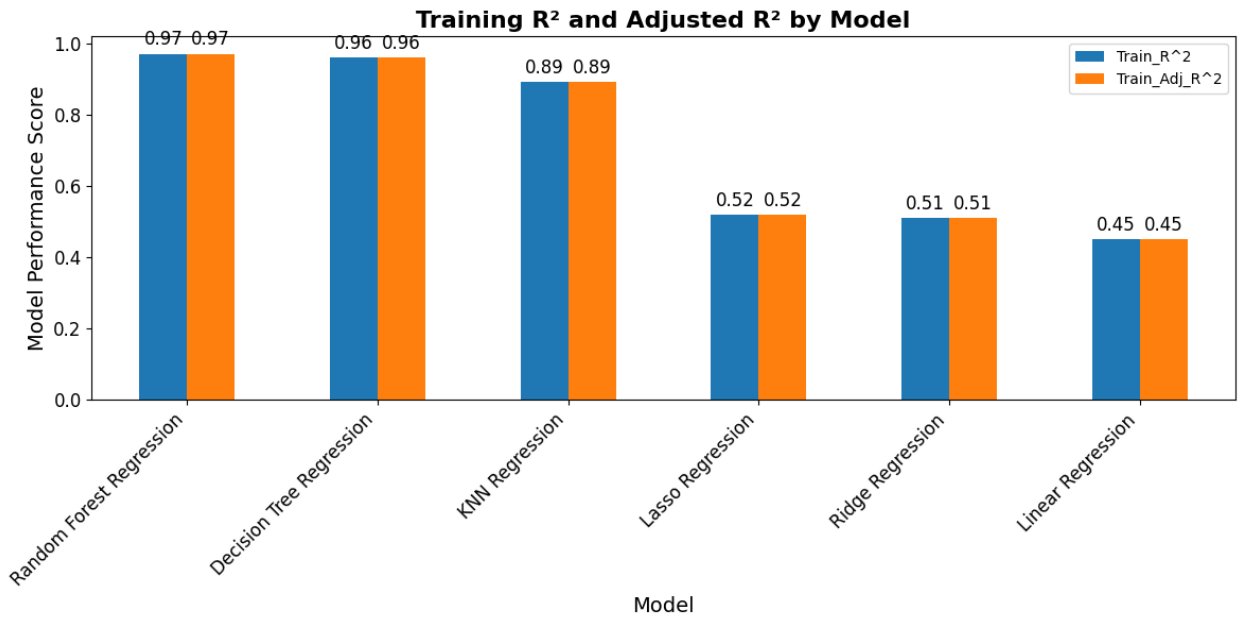
A third brief discussion of the different types of performance metrics is provided before detailing the performance of each proposed model. R-squared indicates how much of the variation in the target variable the model explains. Adjusted R-squared shows how much of the variation in the target the model explains after accounting for the number of predictors, helping prevent overestimating performance when more independent variables are added. The closer the R-squared and Adjusted R-squared values are to 1, the more of the outcome's variance the model captures. It is important to note that it is also ideal for the R-squared and Adjusted R-squared values not to be 1 or too close to 1, as this can indicate overfitting. RMSE measures the average magnitude of prediction errors, with larger errors weighted more heavily. MAE shows the average absolute prediction error in the same units as the target. MAPE shows the average percentage error, making it easy to interpret relative accuracy. For optimal model performance, RMSE, MAE, and MAPE evaluation metrics should be as low as possible.

Fifth, the various models proposed in the milestone analysis showed that linear models underperformed, while nonlinear models excelled. The appendix of this report presents baseline and final model comparisons for each of the six models. To begin with, the Linear Regression, Ridge Regression, and Lasso Regression models could not capture the prominent nonlinear interactions within the dataset. In addition, the KNN Regression geometric-driven model was sensitive to scaling and noise. The two tree-based models, Decision Tree Regression and Random Forest Regression (Random Forest), handled outliers, mixed data types, nonlinear relationships, and interactions among independent variables best in the dataset. Between the Decision Tree Regression and Random Forest Regression models, the Random Forest Regression model performed best (including outperforming the other models) due to the highest R-squared and adjusted R-squared, and the lowest RMSE, MAE, and MAPE. This was due to the model's ability to handle outliers, nonlinear relationships, and mixed types of independent variables, as demonstrated by its ensemble nature. The Random Forest Regression model's performance made it the most dependable contender for deployment. Thus, this aligns with Cars4U's goals of improving accuracy, potency, and generalization across various used car brands, manufacturing years, and conditions.

The six bar charts below compare performance metrics across the six regression models. The first bar chart compares the training R-squared and adjusted R-squared

performance metrics for each of the six models. The second bar chart compares the test R-squared and adjusted R-squared performance metrics for each of the six models. The third bar chart compares the training RMSE and MAE for each of the six models. The fourth bar chart compares the test RMSE and MAE for each of the six models. The fifth bar chart compares the training MAPEs for the six models. The sixth bar chart compares the test MAPEs for the six models. Taken altogether, the Random Forest Regression model is the most accurate and most generalizable; the Decision Tree Regression model is strong but less stable; the KNN Regression model is middle-tier; and the Linear Regression, Ridge Regression, and Lasso Regression models underfit the dataset.

Figure 25. Training R^2 and Adjusted R^2 by Model Bar chart

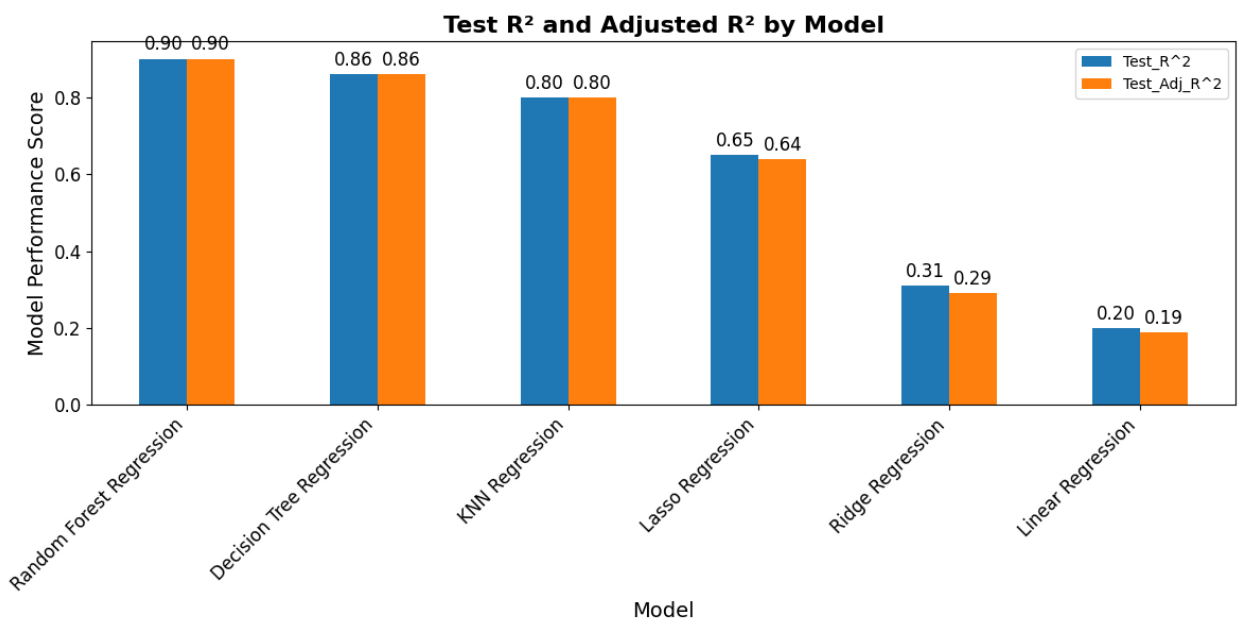


Observations and Insights for Training R^2 and Adjusted R^2 by Model Bar chart

The Decision Tree Regression and Random Forest Regression models have the highest Train R-squared and Adjusted R-squared scores, where they capture complex, nonlinear relationships in the data the best. The Decision Tree Regression model has a Train R-squared, and Train Adjusted R-squared score of 0.96. The Random Forest Regression model has a Train R-squared, and Train Adjusted R-squared score of 0.97. Linear Regression, Ridge, and Lasso have notably lower R-squared and Adjusted R-squared values, indicating that the relationship between the independent variables and Price is not strictly linear. The Linear Regression model has a Train R-squared, and

Train Adjusted R-squared score of 0.45. The Ridge Regression model has a Train R-squared, and Train Adjusted R-squared score of 0.51. The Lasso Regression model has a Train R-squared, and a Train Adjusted R-squared value of 0.52. Linear regularization models improve stability but cannot capture the dataset's nonlinear characteristics. This implies that feature selection and preprocessing were effective. The KNN Regression model has moderate performance but is not competitive; it can capture some nonlinearity but is sensitive to scaling and neighborhood structure. The KNN Regression model has a Train R-squared, and Train Adjusted R-squared score of 0.89. The adjusted R-squared closely tracks the R-squared in each model, indicating that irrelevant predictors do not inflate the models and that most features make meaningful contributions to explaining variance.

Figure 26. Test R^2 and Adjusted R^2 by Model Bar chart

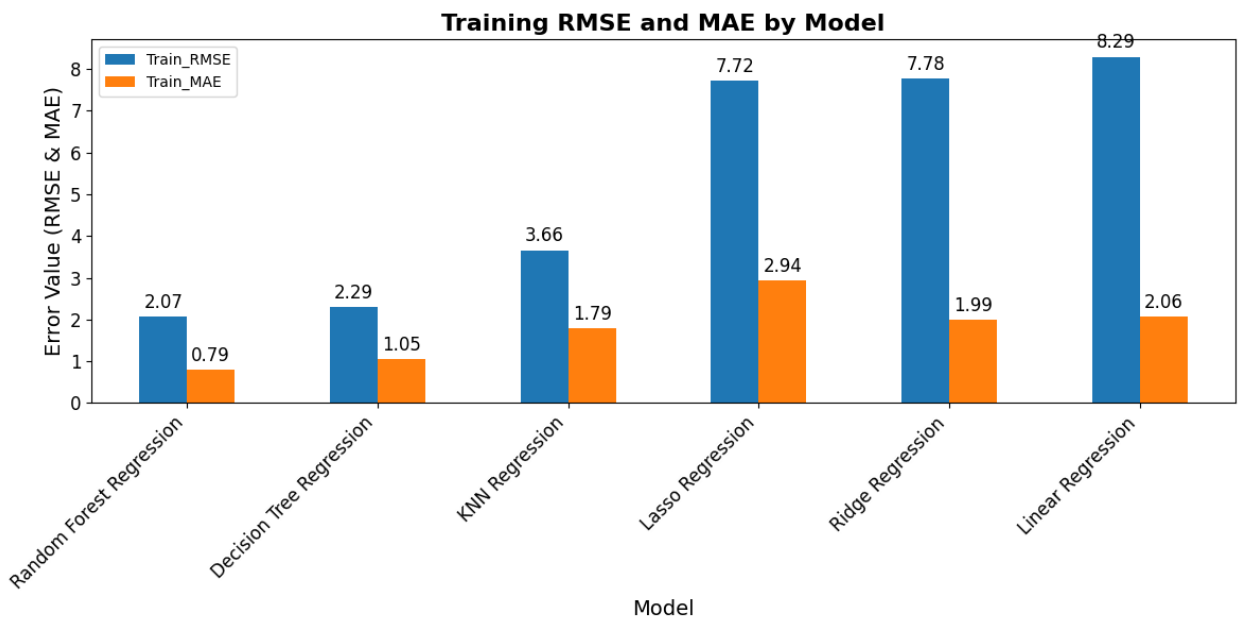


Observations and Insights for Test R^2 and Adjusted R^2 by Model Bar chart

The Decision Tree Regression and Random Forest Regression models have the highest Test R-squared and Test Adjusted R-squared scores, both in the 0.80-0.90 range. In particular, the Decision Tree Regression model has Test R-squared and Test Adjusted R-squared scores of 0.86. The Random Forest Regression model has a Test R-squared and Test Adjusted R-squared score of 0.90. This implies that they capture the nonlinear structure of the data effectively and have the strongest predictive power

on unseen data. Linear Regression, Ridge, and Lasso each have lower Test R-squared values ranging from 0.2 to 0.6. In particular, the Linear Regression model has a Test R-squared of 0.20 and a Test Adjusted R-squared of 0.19. The Ridge Regression model has a Test R-squared score of 0.31 and a Test Adjusted R-squared score of 0.29. The Lasso Regression model has a Test R-squared score of 0.65 and a Test Adjusted R-squared score of 0.64. This implies that linear assumptions do not coincide with the true structure of the dataset. Even with regularization, linear models still cannot capture the interactions and nonlinearities in the dataset. KNN Regression performs moderately, but not competitively, ranking in the middle among all models. The KNN Regression model has Test R-squared and Test Adjusted R-squared scores of 0.80. This implies that KNN can capture some local structure of the dataset but cannot handle high-dimensional, mixed-scale data, and is sensitive to scaling and neighborhood density. The adjusted R-squared closely tracks the R-squared in each model, indicating that irrelevant predictors do not inflate the models and that most features make meaningful contributions to explaining variance.

Figure 27. Training RMSE and MAE by Model Bar chart

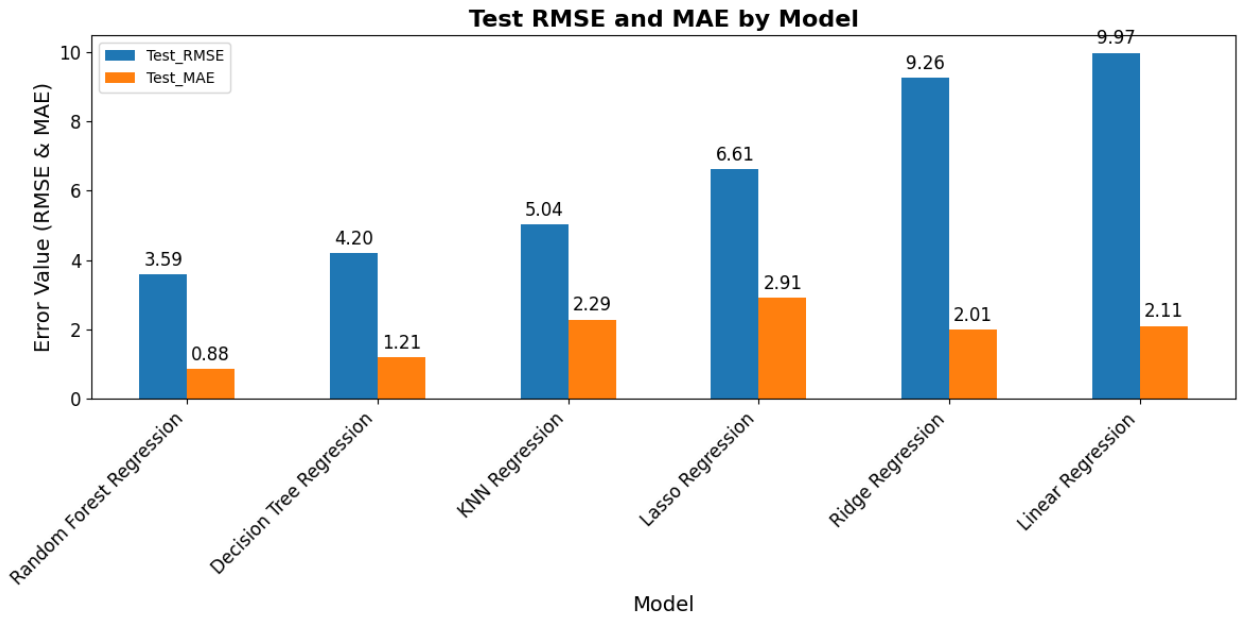


Observations and Insights on Training RMSE and MAE by Model Bar Chart

The Decision Tree Regression and Random Forest Regression models have significantly lower RMSE and MAE, suggesting a tight fit to the training data. The

Decision Tree Regression model has a Train RMSE of 2.29 and a Train MAE of 1.05. The Random Forest Regression model has a Train RMSE of 2.07 and a Test MAE of 0.79. This implies that they can capture nonlinear interactions very well, adapt adequately to the dataset's structure, and may be at risk of overfitting. The Linear Regression, Ridge, and Lasso regression models have notably higher RMSE and MAE than the other models. The Linear Regression model has a Train RMSE of 8.29 and a Train MAE of 2.06. The Ridge Regression model has a Train RMSE of 7.78 and a Train MAE of 1.99. The Lasso Regression model has a Train RMSE of 7.72 and a Train MAE of 2.94. This implies that linear models are underfitting and cannot capture the nonlinear structure of used-car prices. Also, this implies that linear regularization models (e.g., Ridge and Lasso) can stabilize coefficients, but do not improve fitting as much. The KNN Regression model performs moderately well, with lower error than the linear models but higher than the Decision Tree Regression and Random Forest Regression models. The KNN Regression model has a Train RMSE of 3.66 and a Train MAE of 1.79. This implies that the KNN Regression model can capture local patterns but remains sensitive to scaling and neighborhood density. RMSE is invariably higher than MAE across all the models, since RMSE penalizes large errors more significantly, while MAE treats all errors at the same level. This implies that RMSE highlights each model’s ability to handle outliers, and that tree-based models evidently outperform the others.

Figure 28. Test RMSE and MAE by Model Bar chart

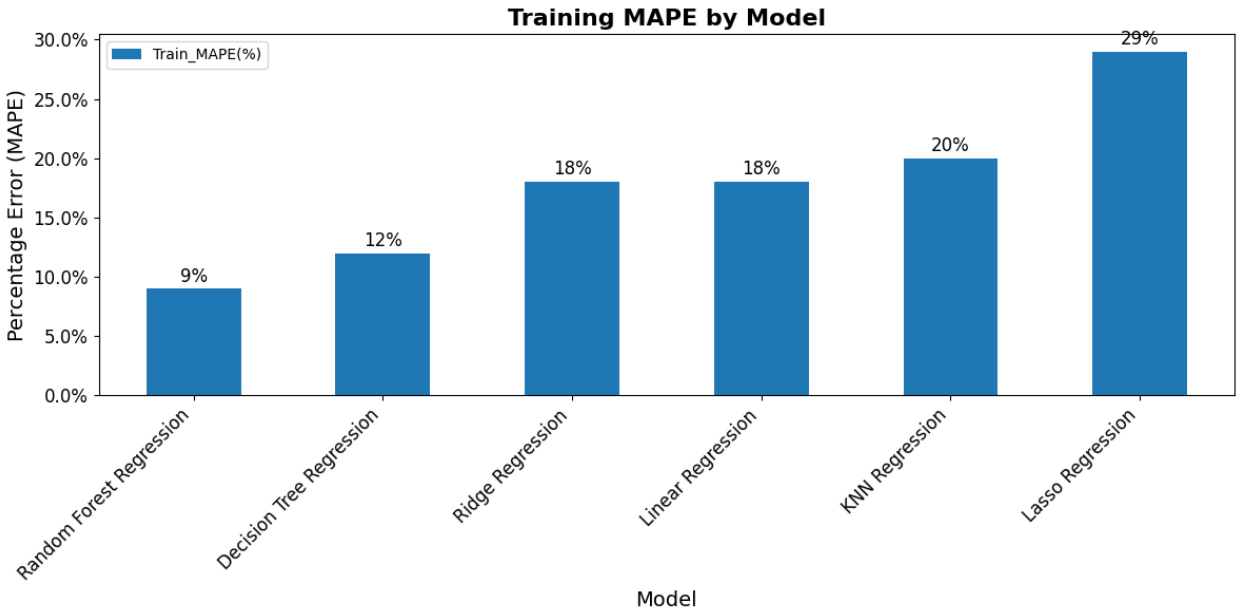


Observations and Insights on Test RMSE and MAE by Model Bar Chart

The Random Forest Regression model has the lowest Test RMSE (3.59) and Test MAE (2.29) across all the models. This implies that the Random Forest Regression model can predict unseen used-car prices with the lowest average error, handle nonlinear relationships and interactions well, and is the strongest candidate for deployment. The Decision Tree Regression model performs well but is slightly worse than the Random Forest Regression model, with a Test RMSE of 4.20 and a Test MAE of 1.21. This implies that the Decision Tree Regression model can capture nonlinear patterns, but it may overfit more than the Random Forest Regression model. The Linear Regression, Ridge Regression, and Lasso Regression models each have very high Test RMSE and Test MAE. The Linear Regression model has a Test RMSE of 9.97 and a Test MAE of 2.11. The Ridge Regression model has a Test RMSE of 9.26 and a Test MAE of 2.01. The Lasso Regression model has a Test RMSE of 6.61 and a Test MAE of 2.91. This implies that they are unable to capture the complicated nature of used-car prices. Even with regularization, linear models cannot capture nonlinear relationships, suggesting they are not the best choice for the prediction task. The KNN Regression model performs moderately well, but is not competitive. The KNN Regression model has a Test RMSE of 5.04 and a Test MAE of 2.29. The KNN Regression model outperforms the linear models, but not as well as the Decision Tree Regression and Random Forest Regression models. This implies that the KNN

Regression model can model local structure, but is also sensitive to scaling and neighborhood density. RMSE is also greater than MAE across all models, as expected, since RMSE penalizes large errors more heavily. This implies that all models exhibit larger deviations, with the Random Forest Regression model reducing them the most.

Figure 29. Training MAPE by Model Bar chart

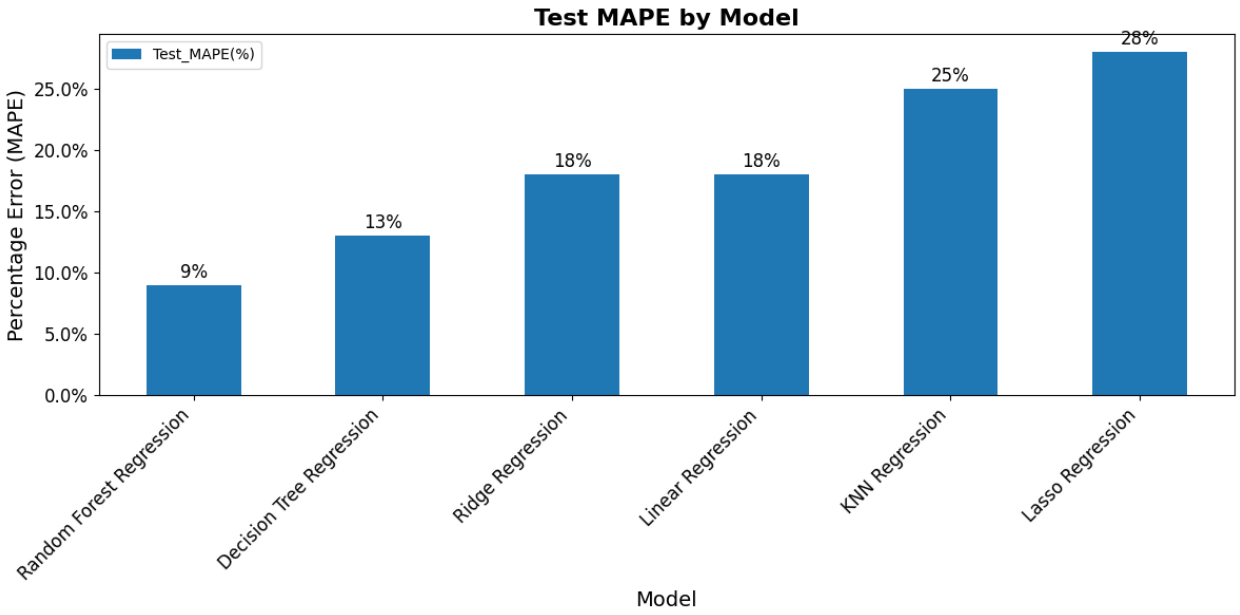


Observations and Insights on Training MAPE by Model Bar chart

The Random Forest Regression model has the lowest training MAPE, with a percentage error of around 9%. This implies that the Random Forest Regression model achieves the highest accuracy on the training dataset, handles nonlinear relationships best, and is the most stable and reliable model. The Decision Tree Regression model has a lower MAPE (12%) than the Random Forest Regression model. This implies that the Decision Tree Regression model captures complex patterns but may overfit more than the Random Forest Regression model. The Lasso Regression model has the highest MAPE (29%). This implies that the Lasso Regression model underfits the data, removes too much signal by aggressively shrinking coefficients, and struggles with nonlinear relationships. The Linear and Ridge Regression models have high MAPEs of 18% each. This implies that linear assumptions do not align with the true structure of used-car prices, and that the Ridge Regression model offers greater stability but not accuracy. The KNN Regression model lies between linear and tree-based models, with

an MAPE of 20%. This implies that the KNN Regression model can capture local patterns but remains sensitive to scaling and neighborhood structure.

Figure 30. Test MAPE by Model Bar chart



Observations and Insights on Test MAPE by Model Bar chart

The Random Forest Regression model has the lowest test MAPE, with the smallest percentage error (9%) on unseen data. This implies that the Random Forest Regression model predicts used-car prices with the highest accuracy, handles nonlinear relationships and interactions the best, and is the strongest model candidate for deployment. The Decision Tree Regression model has a lower MAPE (13%) than the Random Forest Regression model. This implies that the Decision Tree Regression model captures complex patterns but may overfit more than the Random Forest Regression model. The Lasso Regression model has the highest MAPE (28%). This implies that the Lasso Regression model removes too many independent variables due to its aggressive coefficient-shrinking nature, fails to capture nonlinear relationships, and generalizes poorly to unseen data. The Linear and Ridge Regression models have high MAPEs of 18%. This implies that linear assumptions do not align with the true structure of used-car prices, and that the Ridge Regression model offers greater stability but not accuracy. The KNN Regression model falls between linear models and

tree-based models, with a MAPE of 25%. This implies that the KNN Regression model can capture local patterns but remains sensitive to scaling and neighborhood structure.

Sixth, the chosen Random Forest Regression model serves as the final solution design for Cars4U's production pricing engine. The Random Forest Regression model's performance makes it the most reliable candidate for deployment. The model's stability and interpretability are underscored by the importance of independent variables and interpretability tools, which help Cars4U use used-car price valuations for customers and domestic stakeholders. The model aids strategic actions such as dynamic pricing, inventory refinement, and fraud detection. To clarify, the model, as a practical automated pricing engine, enables Cars4U to achieve price standardization, reduce pricing ambiguity, support both buyers and sellers, and establish a competitive advantage in the Indian used-car market. Lastly, the Random Forest Regression model best balances accuracy, robustness, stability, interpretability, and generalization among the proposed models.

Key Takeaways

Big picture, First and all of its Components (that drive the end goal)

The following statements are key takeaways from the Used Cars Price Prediction project. To start with, the central business problem is the underlying price ambiguity in the Indian used-car market. It is critical to note that Cars4U must operate successfully in a used-car market characterized by price unpredictability, information dissymmetry, and erratic pricing. These market properties thus necessitate that Cars4U adopt a machine-learning pricing engine that guarantees equity, clarity, and operational efficiency. Second, the provided dataset is rich but quite inconsistent in quality and requires substantial preprocessing before modeling. The analysis begins with a highly diverse dataset of over 7,253 used cars, which contains important signals but also has considerable shortcomings. These shortcomings include a large number of missing values in the Mileage, Engine, Power, Seats, New price, and Price (target-dependent) variables. Particularly, the new price has over 6,246 missing values. Additional shortcomings include multiple numerical independent variables exhibiting right-skewed distributions due to the presence of luxury vehicles. Other shortcomings include abundant mixed data types requiring encoding and outliers that distort linear

relationships. These conditions validated the need for careful cleaning, imputation, feature engineering, and nonlinear modeling.

Third, exploratory data analysis uncovered vivid economic relationships that drive car prices. Among these relationships are the following powerful and intuitive patterns: higher engine power, engine size, and new-car prices are associated with higher target Price values, higher mileage, older age, and more kilometers driven are associated with lower target Price values, diesel fuel type cars and first-owner vehicles are the most common type of used cars, and luxury used cars comprise the long-tail distributions across multiple variables. These exploratory data analysis insights corroborate the finding that vehicle condition, brand strength, and depreciation behavior are the main independent variables for pricing.

Fourth, feature engineering notably strengthened the modeling foundations for the proposed models. Specifically, the Car Age (derived from Year), Depreciation Amount (which captures brand- and model-level value loss), and Brand (extracted from the Name independent variable) are among the most valuable independent variables. These engineering variables improved interpretability and helped models better capture real-world pricing logic.

Fifth, linear and geometric regression models were inadequate, whereas nonlinear tree-based regression models better captured the Indian used-car market. To be specific, Multiple Linear Regression, Ridge Regression, and Lasso Regression had difficulty with nonlinear interactions. KNN Regression was highly sensitive to scaling and noise. The tree-based Decision Tree Regression and Random Forest Regression models both performed far better than all other models. Between these two tree-based models, Random Forest Regression outperformed Decision Tree Regression due to its better handling of outliers, nonlinear relationships, mixed categorical and numerical variables, and complex interactions among independent variables. At this point, it was evident that the dataset and market structure of the Indian used-car market necessitated nonlinear regression, particularly tree-based methods.

Sixth, as noted earlier, the Random Forest Regression model outperformed all other regression models, making it the strongest and most dependable model. Across the R-squared, Adjusted R-squared, RMSE, MAE, and MAPE performance metrics, the Random Forest Regression model outperformed all other models for each performance

metric. Hence, the Random Forest Regression model is accurate, robust, generalizable, business-aligned, and interpretable through feature (independent variable) importance. Overall, the Random Forest Regression model best aligns with Cars4U's goals: standardized pricing, reduced negotiation friction, and adaptable automation. Finally, the Random Forest Regression model demonstrates a robust foundation for deployment, giving clear next steps for enhancement and scaling.

Summary of the most important findings

To put it all together, the final six key takeaways or important findings can be summarized. Cars4U requires a machine-learning pricing engine to address market instability in the Indian used-car market. The initial dataset provided is eclectic and noisy, requiring thorough preprocessing and feature engineering. The Exploratory Data Analysis uncovered clear, economically meaningful independent variables (predictors) of used-car prices. The nonlinear models outperform the linear models, which illustrates the complexity of the pricing problem. The Random Forest Regression model is the best-performing, offering high accuracy, robustness, and interpretability. The previous milestone report underscores a strong foundation for deploying the Random Forest Regression model, with clear next steps for further advancement and scaling.

Key Next Steps

Steps Taken to Improve the Solution and Make the Best of the Solution

The following statements are the key next steps for the Used Cars Price Prediction project. These steps center on improving the solution, optimizing business value, and indicating what stakeholders should do next to move toward deployment. First, the highest-priority next step is to strengthen data quality and enrich the dataset. The rationale behind this step is that the dataset is powerful, but not complete. Consequently, the next steps should primarily focus on improving the reliability of the inputs feeding the model. Several actions, such as imputing or replacing missing values for New Price (6,247 missing), Power (175 missing), Engine (46 missing), Seats (53 missing), Mileage (83 missing), and Price (1,234 missing) in a more substantial manner, obtain additional data sources, such as real-world transaction prices, market demand benchmarks, used car condition reports, and seller behavior patterns, and standardize

categorical fields (e.g., inconsistent naming in Name or Location). This step matters because better data yields a stronger signal, which in turn leads to higher model accuracy and stability.

Second, the next key step is to expand and refine the feature engineering independent variables by adding more meaningful independent variables. Among these are adding interaction independent variables (e.g., Power X Age–how the value or performance impact of a car's power changes as the car gets older, and Brand X Fuel Type–the impact of a specific brand on the price outcome depends on the fuel type the car uses). Other feature-engineered independent variables include geographic pricing indicators (e.g., city-level demand or supply), condition-based independent variables (e.g., service history and accident reports), and improved depreciation modeling that uses nonlinear transformations. The rationale behind this step is that feature engineering is the largest performance driver for table-generated machine learning problems.

Third, the next key step is to further optimize and validate the Random Forest Model. First, it would be ideal to perform hyperparameter tuning (e.g., grid search or Bayesian optimization). Second, we should evaluate cross-validation to ensure generalization across used-car segments. Third, it would be ideal to compare the Random Forest Regression model with other nonlinear regression models, such as Gradient Boosting Machines (e.g., XGBoost, LightGBM, and CatBoost), as well as stacking or ensemble-blended models. Fourth, we should perform further error analysis to identify which brands or segments are the most difficult to predict, whether luxury used-car vehicles disrupt predictions, and whether older used-car vehicles require separate modeling. The rationale for this step is that the regression model must be robust across all used-car vehicle types to support real-world deployment.

Fourth, the next key step is to establish tools for explainability and transparency to build trust between buyers and sellers. To start with, it would be ideal to incorporate feature importance (i.e., independent variable importance) dashboards through using tools such as SHAP (SHapley Additive exPlanations–a game-theory-based method used in data science to interpret machine learning models by measuring the contribution of each feature to a prediction), permutation importance. Secondly, it would be ideal to provide price justification summaries for each prediction. Third, it

would be ideal to formulate confidence intervals or uncertainty estimates for predicted target prices. The rationale behind this step is that explainability lowers negotiation friction and improves user trust, both of which are core business goals.

Fifth, the next key step is to prepare future models, beyond the final chosen Random Forest Regression model, for real-world deployment, since Cars4U requires an adaptable, scalable, automated pricing engine. First, a model inference pipeline should be developed for batch or real-time inference. Second, it is evident that data refresh cycles should be set up weekly or monthly. Third, it is evident that model retraining schedules should be implemented to adapt to market changes. Fourth, it is evident that the model should be implemented in Cars4U's seller onboarding workflow, buyer browsing experience, and internal valuation tools. The rationale behind this step is that a finalized model is only invaluable once it is operational, stable, and ongoingly updated.

Steps to be followed by stakeholders

As for the key next stakeholder-specific next steps, the following must be adhered to across the data science team, the engineering or platform team, the product and business teams, and the leadership teams. For the data science team, the team must enhance data quality and feature engineering, tune and validate the Random Forest Regression model, establish explainability tools, conduct regular error analysis, and perform segment-level performance checks. The engineering or platform team should prepare a framework for model deployment, construct APIs or batch pipelines for price predictions, and ensure data ingestion pipelines are balanced and automated. For the product and business teams, the teams must carefully define how predicted prices will be displayed to users, determine acceptable error margins for various car categories, and identify business rules for price overrides, confidence thresholds, and handling of outlier used-car vehicles. For the leadership team, the team must approve appropriate data-acquisition budgets, align the pricing engine model with Cars4U's market-augmentation strategy, and set Key Performance Indicators (KPIs) such as reducing negotiation time, increasing seller conversions, and improving buyer trust metrics.

Altogether, Cars4U should adhere to the following five highlights to make the best use of the solution. First, Cars4U should utilize the model to standardize pricing

across geographic locations to reduce negotiation friction. Second, Cars4U should gain insights into the importance of features (independent variables) to inform procurement and inventory decision-making. Third, Cars4U should use predicted prices to identify underestimated used cars to improve its margins. Fourth, Cars4U should deploy the final Random Forest Regression model as a customer-facing tool to create trust and clarity among its buyers and sellers. Lastly, Cars4U should readily retrain the chosen Random Forest Regression model as new data arrives to maintain accuracy.

Final Proposed Model Specifications (Proposed Solution Design)

The Random Forest Regression model is the final adopted solution for Cars4U's pricing engine. It provides the best predictive performance, the highest generalization capability, and the most stabilized error behavior among all six evaluated regression models. In general, the Random Forest Regression model outperforms all other models, making it the most reliable candidate for deployment.

Model Type

The Random Forest Regression model's preliminary characteristics are the following. The proper name for this model or algorithm is Random Forest Regression. It belongs to the family of ensemble tree-based models. An ensemble model family is a group of machine learning methods that combine individual models to produce a stronger, more robust overall prediction. The idea is that single, diverse models make various errors, but combining them reduces variance, bias, or both. In effect, ensemble methods frequently outperform single models because they can capture a richer set of patterns in the data. The training target variable is the log-transformed Price in INR, scaled by 100,000. The prediction output is defined as the exponentiation of the training target variable back to the original INR price scale.

Model Configuration and Hyperparameters

The final Random Forest Regression model is a hyperparameter-tuned version with the following properties. It has a maximum depth of 5, which was chosen to reduce overfitting while preserving a nonlinear structure. Max depth is the maximum number of nodes across all decision trees in the Random Regression model, expanding until all leaves are pure or contain fewer than the minimum samples required for a split. Minimum samples split is a hyperparameter that specifies the minimum number

of samples required at an internal node for it to be considered for further splitting. If a node has fewer samples than the minimum number of samples required for splitting, the splitting procedure halts for that node, and it becomes a leaf node. The criterion is squared error with a minimum sample leaf value of 1. The squared error criterion measures split quality by computing the average squared difference between predicted and actual target values, aiming to minimize the loss (variance). Specifically, it computes the average of squared residuals to evaluate model accuracy. The minimum sample leaf parameter in a Random Forest Regression model indicates the minimum number of samples needed to exist in a leaf (terminal) node. It is mainly used to manage the complexity of trees (decision trees) in the forest and to avoid overfitting.

In addition, the final chosen hyperparameter-tuned Random Forest Regression model has the additional properties. It has a bootstrap-enabled component and a random state of 1 component to ensure reproducibility. A bootstrap-enabled component, also known as bagging, produces diverse training sets for each decision tree in the Random Forest Regression model. It ensures that each decision tree is trained on a distinct, randomly resampled subset of the original data (via sampling with replacement) to reduce correlation among the trees and improve model robustness. A random state of 1 secures the seed for the pseudo-random number generator, ensuring the model produces identical outputs every time it is generated. Thus, it specifies the reproducibility of bootstrap sampling and of independent variable (feature) selection during node splitting, ensuring uniformity across experiments. The independent variable inputs are 43 distinct engineered and encoded after fully preprocessing the dataset. They are found in the table below. Moreover, using a 70-30 train-test split, the data will be divided into training and test sets, with 70% reserved for training and 30% for testing, given the dataset's 6,018 rows. After splitting the data, the initial training set contains 4,212 rows and 43 columns, and the test set contains 1,806 rows and 43 columns.

Table 1. Initial Independent Variables (Features)

```
<class 'pandas.core.frame.DataFrame'>
Index: 4212 entries, 2116 to 5158
Data columns (total 43 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Year                       4212 non-null   int64
 1   Mileage                    4212 non-null   float64
 2   Engine                     4212 non-null   float64
```

```
 3   Power                      4212 non-null   float64
 4   New_price                  4212 non-null   float64
 5   Kilometers_Driven_log      4212 non-null   float64
 6   Car_Age                    4212 non-null   int64
 7   Depreciation_Amount        4212 non-null   float64
 8   Location_Bangalore         4212 non-null   int64
 9   Location_Chennai           4212 non-null   int64
 10  Location_Coimbatore        4212 non-null   int64
 11  Location_Delhi             4212 non-null   int64
 12  Location_Hyderabad         4212 non-null   int64
 13  Location_Jaipur            4212 non-null   int64
 14  Location_Kochi             4212 non-null   int64
 15  Location_Kolkata           4212 non-null   int64
 16  Location_Mumbai            4212 non-null   int64
 17  Location_Pune              4212 non-null   int64
 18  Fuel_Type_Other_Fuel_Type  4212 non-null   int64
 19  Fuel_Type_Petrol           4212 non-null   int64
 20  Transmission_Manual        4212 non-null   int64
 21  Owner_Type_Owner_3plus     4212 non-null   int64
 22  Owner_Type_Second          4212 non-null   int64
 23  Seats_5.0                  4212 non-null   int64
 24  Seats_7.0                  4212 non-null   int64
 25  Seats_8.0                  4212 non-null   int64
 26  Seats_Rare_Seat_Values     4212 non-null   int64
 27  Brand_audi                 4212 non-null   int64
 28  Brand_bmw                  4212 non-null   int64
 29  Brand_chevrolet            4212 non-null   int64
 30  Brand_ford                 4212 non-null   int64
 31  Brand_honda                4212 non-null   int64
 32  Brand_hyundai              4212 non-null   int64
 33  Brand_land                 4212 non-null   int64
 34  Brand_mahindra             4212 non-null   int64
 35  Brand_maruti               4212 non-null   int64
 36  Brand_mercedes-benz        4212 non-null   int64
 37  Brand_nissan               4212 non-null   int64
 38  Brand_renault              4212 non-null   int64
 39  Brand_skoda                4212 non-null   int64
 40  Brand_tata                 4212 non-null   int64
 41  Brand_toyota               4212 non-null   int64
 42  Brand_volkswagen           4212 non-null   int64
dtypes: float64(6), int64(37)
memory usage: 1.4 MB
```

Final Model Performance

To reiterate, the final Random Forest Regression model performs well, as indicated by R-squared values of 0.97 on the training dataset and 0.90 on the test dataset. The performance results show that it is generalizing quite well. The slight increase in RMSE from 2.07 in the training dataset to 3.59 in the test dataset suggests some overfitting, although the gap is modest and normal for an ensemble machine learning model. MAE is low on both the training and test datasets, and the identical train and test MAPE values of 0.09 suggest exceptionally stable percentage-based

accuracy in the test dataset. Overall, this Random Forest Regression model has a healthy balance between accuracy and generalization. This makes the Random Forest a dependable choice for prediction. Most importantly, this Random Forest Regression model is preferred for its greater interpretability and tendency to avoid overfitting the training dataset. The performance metrics for the Random Forest Regression model are shown in the table below.

Table 2. Final Hyperparameter-Tuned Random Forest Regression Model Performance Metrics

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.97	0.90	0.97	0.90	2.07	3.59	0.79	0.88	9%	9%

Most Important Independent Variables (Features)

Independent variable (feature) importance measures how much each independent variable contributes to enhancing the model's predictions. In both Decision Tree Regression and Random Forest Regression models, feature importance is measured by how much a feature reduces prediction error when used to split the data. Higher importance scores indicate features that play a larger role in explaining the target variable. While Decision Tree Regression importance values are computed from a single Decision Tree, Random Forest Regression importance values represent the average contribution of each feature across all individual Decision Trees in the model.

To establish a clear, interpretable baseline, the Decision Tree feature-importance plot is presented first in the bar chart below. This bar chart provides an intuitive view of the same key features as the following complex, higher-performing Random Forest Regression model. The Random Forest feature-importance plot is presented second in the bar chart below the Decision Tree feature-importance plot.

Figure 31. Decision Tree: Top 3 Feature Importances Bar chart

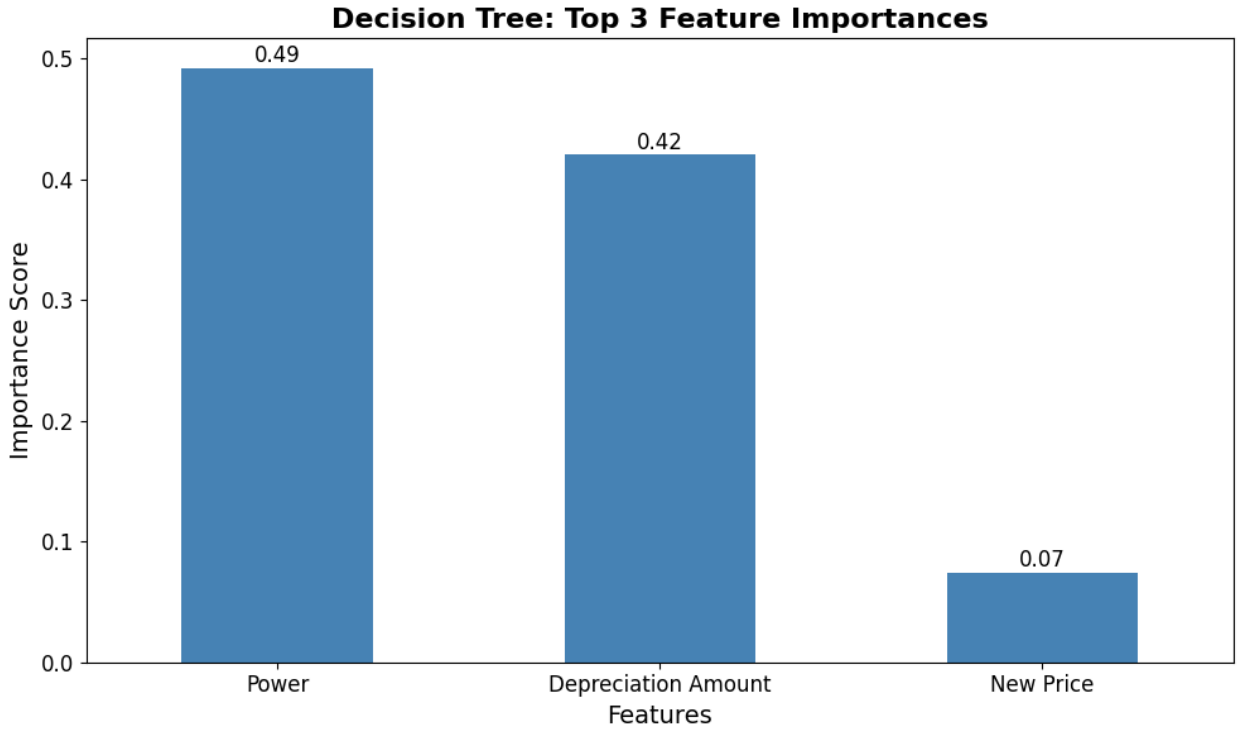
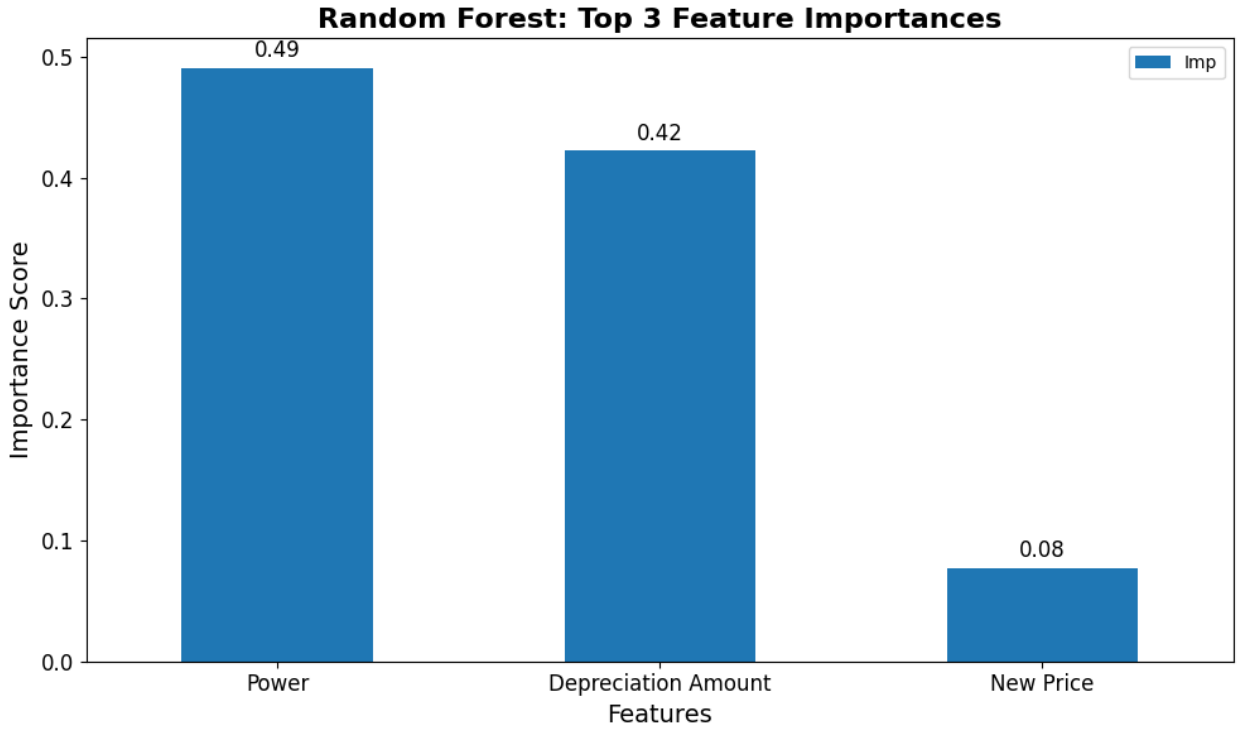


Figure 32. Random Forest: Top 3 Feature Importances Bar chart



Observations and Insights on Decision Tree: Top 3 Feature Importances and Random Forest: Top 3 Feature Importances Bar charts

The most important independent variables (features) from the Decision Tree Regression and Random Forest Regression models below form a concentrated set of predominant features. These include Power, Depreciation Amount, and New Price. Power and Depreciation Amount dominate the model's overall predictive signal. New Price also contributes purposefully, but is still far behind Power and Depreciation Amount. In particular, this suggests that temporal and usage-based effects contribute little to the importance of independent variables after Power and Depreciation Amount. Overall, these independent variables align with realistic automotive economics, in which performance, value retention, and original market value shape used-car price valuations.

Data Inputs, Data Cleaning, and Preprocessing Requirements

The Random Forest Regression model employs a fully engineered and cleaned dataset of 6,018 rows with 43 independent variables, including the following: A Car Age independent variable, where it is computed as subtracting each used car's manufacturing year from the reference year, which in this case is the maximum Year value from the Year independent variable, or 2019. A Depreciation Amount independent variable is calculated as the subtraction of the target Price variable from the New Price variable. A Log(Kilometers Driven) transformed independent variable, and Log(Price) transformed dependent variable are important feature-engineered numerical variables. Given the importance of the original Kilometers Driven independent variable (i.e., feature), it must be log-transformed to reduce or eliminate its skewness. Other reasons for this log transformation include that Kilometers Driven spans several orders of magnitude and has a non-linear, diminishing-returns relationship with the target price variable. Also, applying a log transformation reduces the influence of cars with extremely high mileage, improves the linearity of the relationship with the target price variable, and enhances interpretability. Because the original Price variable is the target variable, it must be log-transformed to reduce skewness and improve model stability. Also, the distribution of the Log(Price) variable becomes more symmetrical, with reduced skewness and far fewer outliers. The Serial Number variable was initially dropped from the dataset because it is an identifier with

no predictive value. Brand and Model are extracted from the original Name (the used car's name, including Brand and Model) and treated as independent categorical variables. The grouping of rare categories in the Fuel Type, Owner Type, Seats, and Brand categories is also used to stabilize linear models and avoid sparse-category issues. Lastly, no outliers were capped or removed. All outlier values are true, real, and meaningful, reflecting genuine used-car characteristics. Capping or removing outliers would instead distort the distributions of these variables, introduce artificial sanitization bias, reduce model transferability, and remove significant correlation patterns. Specifically, most outliers in the dataset represent actual premium used cars. In addition, when outliers are capped in each numerical independent variable, each proposed model would be weakened in its ability to capture price differences. Moreover, it would reduce variance, degrade predictive performance, and misinterpret important numerical variables.

There are also imputation strategies that were executed to fill in the missing values for the New price, Seats, Engine, Power, and Mileage independent variables. There are 83 missing values for Mileage (now including the data entry error values in the variable with values of 0.0), 46 missing values for Engine, 175 missing values for Power, 53 missing values for Seats, 6,246 missing values for New price, 1,234 missing values for Price, and 1,234 missing values for Price log. Since there are 6,246 missing values for the New price variable, a hierarchical imputation strategy is implemented to avoid introducing noise into the variable. To clarify, the imputation strategy involves three-level hierarchical steps. The first step involves imputing the median price for the same Brand and Model combination of a used car using a best-guess procedure. The second step involves imputing the median for each brand as a second-best guess. The third final step involves imputing the overall median New price of the entire dataset. Similarly, a hierarchical imputation strategy is used for the Seats variable. To be specific, the imputation strategy uses a two-step hierarchical approach that relies on the typical seating capacity of similar used cars. First, the median for each brand-model combination is calculated and assigned back to the corresponding rows. This step allows most missing entries to be filled in using the seating arrangement for that specific type of used car. The second step involves falling back to the overall median of the Seats column when no brand-model combination is available. This ensures that every missing value is filled in using the most definitive and appropriate information available. For imputing missing values for Mileage, Engine, and Power, the median of

each variable is used. Unlike other independent variables, the median will be used to fill in the mileage values initially set to 0.0, except for the number of missing values indicated in the table below. Since 1,234 rows contain missing values in the target Price and transformed target Price variables, these rows cannot be used for modeling and will be removed. The rationale for dropping rows is to avoid introducing bias into the data. It is important to clarify that the Depreciation Amount independent variable was not feature-engineered until the missing values in the New Price independent variable were imputed and the missing values in the target Price dependent variable were dropped. The table below presents an updated summary of missing values by variable.

Table 3. Updated Missing Value Summary by Variable

Name	0
Location	0
Year	0
Kilometers_Driven	0
Fuel_Type	0
Transmission	0
Owner_Type	0
Mileage	83
Engine	46
Power	175
Seats	53
New_price	6246
Price	1234
Kilometers_Driven_log	0
Price_log	1234
Brand	0
Model	0
Car_Age	0

Reasons why the Random Forest Regression Model is the Final Choice

The Random Forest Regression model is chosen as the final model because of the following reasons. It best captures nonlinear relationships and interaction-heavy variables, notably found in the Indian used-car market. It also handles mixed data

types best, especially continuous variables, one-hot-encoded categorical variables, and engineered variables. It is also not sensitive to outliers or noise, so that it can capture luxury-segmented used cars as real market signals rather than errors. Also, it achieves the highest R-squared on the test set, at 0.90, the highest among all models. Most importantly, the model provides interpretable importance scores for independent variables, which support business clarity and customer trust.

Reasons for Production-Readiness

The Random Forest Regression model is considered production-ready for the following reasons. Foremost, it has high accuracy and low error across all performance metrics. Secondly, it is stable across train-test splits (70-30), with 70% of the dataset used for training and 30% for testing. Thirdly, it supports real-time inference due to the speed of tree-ensemble predictions. Fourthly, it provides strong explanatory power through feature-importance and SHAP-based tools. Fifthly, it aligns with Cars4U's operational needs for dynamic pricing, inventory optimization, and fraud detection. Given these reasons, the Random Forest Regression model is viable for production deployment and offers sufficient interpretability for business use.

Model Limitations and Tradeoffs

The Random Forest Regression model has the following limitations and tradeoffs. First, it has a higher computational cost than linear models. Specifically, this is due to longer training times, higher memory usage, and more costly hyperparameter tuning. Second, it has lower interpretability than linear regression models (e.g., Multiple Linear Regression, Ridge Regression, and Lasso Regression). Third, it is more sensitive to noisy or irrelevant dummy variables. Fourth, it is more complex to deploy and monitor because of its larger model size, which comprises many decision trees. The Random Forest Regression model is considerably larger, which can be an issue on some resource-limited systems or devices. Overall, the Random Forest Regression model has a heftier computational cost, reduced interpretability, and is more complicated to deploy and monitor.

Problem and Solution Summary

Problem Summary

The Indian used-car market is a rapidly growing segment of the automotive industry, characterized by price unpredictability, geographic disparities, and information asymmetry between buyers and sellers. Unlike new cars, used cars do not have standardized pricing. Thus, this leads to negotiation friction, inconsistent estimations, and decreased trust across used-car platforms. This atmosphere also entails numerous buyers and sellers operating with incomplete information and notable regional disparities. Nevertheless, the entire Indian used-car market relies on authentic, reliable, and transparent pricing to address the aforementioned unfavorable characteristics. These inherent standardization deficiencies in the Indian used-car market can affect used-car platforms' ability to compete with other platforms.

Cars4U, a used-car startup company, intends to enter the used-car market in India. The main objective of this company is to provide accurate, unbiased, and scalable price estimates that enhance clarity, reduce negotiation friction, and improve operational productivity. The principal mission of this company is to develop a machine-learning-driven pricing engine model that brings fairness, clarity, and scalability to every used car in Cars4U's inventory. Given the previously stated characteristics of the Indian used-car market, the machine-learning-driven pricing engine must generate accurate, reliable, and scalable pricing for each used car while avoiding bias, price instability, and information discrepancies. A robust predictive model can significantly enhance Cars4U's market position by reducing negotiation disputes. This results in increased trust between buyers and sellers and enables the company to implement automatic, data-driven pricing. Therefore, the machine-learning pricing engine would empower Cars4U to provide clear valuations, reduce negotiation conflicts, improve user confidence, and establish an adaptable, data-driven pricing strategy that strengthens long-term competitive advantage. As a result, the specific problem Cars4U aims to solve is the lack of a reliable, data-driven method to predict unbiased used-car prices in the Indian market, particularly relative to competitors. Overall, the specific problem this project aims to address is the lack of a reliable, data-driven method for predicting fair used-car prices in the Indian market.

Final Proposed Solution Design

The proposed solution design is driven by Cars4U's mandate to replace India's erratic, intuition-driven used-car pricing landscape with a standardized, data-driven,

scalable valuation system. Due to the nature of the Indian used car market being inconsistent, opaque, and an extremely variable pricing environment where multiple buyers and sellers operate with incomplete information and noteworthy regional differences, Cars4U needs a pricing engine regression model that can lower bias, remove information asymmetry, and produce pricing valuations across brands, cities, and vehicle conditions. Therefore, the solution design is built around a machine-learning model capable of capturing nonlinear interactions that drive used-car prices, which humans cannot reliably evaluate. Notably, brand reputation, geographic demand, past ownership history, and vehicle condition interact nonlinearly. A durable machine learning model is essential for standardizing pricing, supporting operational decisions, and enhancing Cars4U's competitive position in the Indian used-car market.

Key Points that Describe the Final Proposed Solution Design

The following are six key points that describe the final proposed solution design. First, a comprehensive data-driven methodology must be adopted as part of the final proposed solution design. As such, the solution uses a structured machine learning approach that includes data cleaning, handling missing values, extracting independent variables (features), performing feature engineering, and encoding categorical variables to prepare the data for modeling. Second, an evaluation of multiple modeling techniques is required as part of the final proposed solution design. Cars4U tested a Multiple Linear Regression model, regularized linear regression models (e.g., Ridge Regression and Lasso Regression), a geometric-based K-nearest neighbors (KNN) Regression model, and nonlinear tree-based models (e.g., Decision Tree Regression and Random Forest Regression) to identify the top-performing regression model suitable for the pricing engine model. The table below shows the performance across each model.

Table 4. Performance Metrics of Final Regression Models

	Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
Linear Regression	0.45	0.20	0.45	0.19	8.29	9.97	2.06	2.11	18%	18%

	Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
Ridge Regression	0.51	0.31	0.51	0.29	7.78	9.26	1.99	2.01	18%	18%
Lasso Regression	0.52	0.65	0.52	0.64	7.72	6.61	2.94	2.91	29%	28%
Decision Tree Regression	0.96	0.86	0.96	0.86	2.29	4.20	1.05	1.21	12%	13%
Random Forest Regression	0.97	0.90	0.97	0.90	2.07	3.59	0.79	0.88	9%	9%
KNN Regression	0.89	0.80	0.89	0.80	3.66	5.04	1.79	2.29	20%	25%

Performance of Different Models (Observations and Insights)

As six distinct models are used to address the business problem of predicting the price of a used car at INR 100,000, as requested by the Cars4U startup, each yields unique performance metrics.

To begin with, the final chosen Multiple Linear Regression model has the following performance metrics highlighted. It has an R-squared of 0.45 on the training dataset and 0.20 on the test dataset, with high RMSE values ranging from 8.29 to 9.97. These highlighted performance results suggest that a Multiple Linear Regression model underfits and poorly generalizes unseen data. Consequently, a Multiple Linear Regression model is not ideal for capturing the overall structure of the dataset. Furthermore, Ridge and Lasso Regression models, similar in structure to Multiple Linear Regression, show some stability improvements over Multiple Linear Regression but remain limited under linear assumptions. Regularization in both Ridge and Lasso

Regression mildly improves model performance but is unable to capture nonlinear relationships in the target variable, the used-car price. The overall performance of both the Ridge and Lasso Regression models shows slight improvements, but tree-based models still outperform them.

In addition to linear regression models, Decision Tree Regression and Random Forest Regression were chosen as alternative tree-based models, and a KNN regression model was chosen as a non-parametric, geometric-based model. The final tuned Decision Tree Regression model has an R-squared of 0.96 on the training dataset and 0.86 on the test dataset. The Decision Tree Regression model also has low MAE and MAPE values, with some overfitting but robust generalization. It can be viewed as a preliminary baseline tree-based model because it efficiently captures nonlinear interactions. The final chosen Random Forest Regression-tuned model has an R-squared of 0.97 on the training dataset and 0.90 on the test dataset. The Random Forest Regression model also has very low RMSE, MAE, and MAPE. Compared to the Decision Tree Regression model, the Random Forest Regression model offers higher accuracy, more stable generalization, and more dependable predictive performance. The final KNN Regression model has an R-squared of 0.89 on the training dataset and 0.80 on the test dataset. The KNN Regression model has modest RMSE and MAE values, is sensitive to local noise, and performs fairly well but is not as effective as the final chosen Random Forest Regression model.

Third, in addition to using a comprehensive, data-driven methodology and evaluating multiple modeling techniques, the Random Forest Regression model was selected as the final model. Reasons for selecting the Random Forest Regression model include its ability to capture nonlinear relationships best and deliver superior accuracy. As noted in the table above detailing the performance metrics of each final chosen model, the Random Forest Regression model had the highest R-squared and adjusted R-squared scores and the lowest RMSE, MAE, and MAPE scores. These performance metrics were achieved because the model can handle outliers, nonlinear relationships, and mixed types of independent variables. Fourth, the final solution design emphasizes interpretability and the importance of independent variables (features). The final proposed solution design includes interpretability tools to assist Cars4U in validating pricing valuations. Hence, the model must be stable and

interpretable, with the significance of independent variables underscoring both internal and customer-facing explanations.

Fifth, the final proposed solution design requires a deployment-ready, scalable pricing engine. To clarify, the Random Forest Regression model is implemented as a production-ready engine that supports automation, dynamic pricing, and operational decision-making. Accordingly, the Random Forest Regression model predicts fair market values and provides a scalable, logical foundation for Cars4U's pricing automation, inventory enhancements, and long-term competitive strategies. Sixth, the final proposed solution design requires alignment with Cars4U's strategic business initiatives. The solution must be designed to predict used-car prices, enhance customer trust, reduce negotiation traction, and improve Cars4U's market position. For this reason, the final proposed model would enable Cars4U to achieve price standardization, reduce pricing ambiguity, benefit buyers and sellers, and build competitive advantages in the Indian used-car market.

How the Final Solution Design Model is Likely to Solve the Problem

The final solution design, the Random Forest Regression model, is likely to solve the business problem in the following three ways. To start with, the final solution design model captures the nonlinear, intricate structure of used-car pricing. Specifically, it handles nonlinear relationships, mixed variable types, outliers, and interaction effects best. Hence, the model best captures the actual dynamics of used-car pricing in the Indian used-car market. Secondly, the Random Forest Regression model had the best empirical performance across R-squared, Adjusted R-squared, RMSE, MAE, and MAPE metrics compared to all other tested models. In other words, the Random Forest Regression model is the most accurate and dependable model for Cars4U to deploy. Thirdly, the Random Forest Regression Model is in alignment with Cars4U's functional and deliberate goals. These goals include improving accuracy, potency, and generalization across used car brands, manufacturing years, and condition. Consequently, the solution is both strategically appropriate and technically sound.

Reasoning Behind the Proposed Solution Design Model

The final solution, the Random Forest Regression model, is the optimal proposed solution in the following three ways. First, the Random Forest Regression model captures the complex intricacies of the Indian used-car market, where used-car attributes and factors such as brand reputation, engine size, power, age, mileage, and location interact nonlinearly. In simpler terms, the Random Forest Regression model employs a data-driven approach to model realistic real-world behavior in the Indian used-car market. Second, the Random Forest Regression model is very robust to disordered, noisy data, especially given the dataset's skewed distributions, outliers, missing values, and mixed categorical and numerical variables. Therefore, the Random Forest Regression model is the most viable choice. Lastly, the Random Forest Regression model is interpretable for business stakeholders, with both durability and explainability supported by the necessity of independent variables and the interpretability tools used. For this reason, the Random Forest Regression model permits Cars4U to explain prices to customers and internal team stakeholders, making it the best model to adopt.

How the Final Solution Design Model Affects the Business

The final solution design, the Random Forest Regression model, affects the Cars4U business in the following four ways. To begin with, the Random Forest Regression model empowers Cars4U to have fair, clear, and standardized pricing. To clarify, the Random Forest Regression model helps Cars4U address the main business problem, where pricing inconsistency and information disparity are the norm in the Indian used-car market. At the same time, the Random Forest Regression model helps Cars4U reduce price opacity for both buyers and sellers, an advantage over other platforms. Additionally, the Random Forest Regression model lessens negotiation adversity and promotes trust. In fact, the Random Forest Regression model would lessen negotiation conflicts, which would yield in augmented trust between buyers and sellers. Due to the nature of the Indian used-car market, trust is a vital differentiator. Likewise, the Random Forest Regression model would best support inventory enhancement and revenue growth, with profit-driving capacity being integrated. Notably, the Random Forest Regression model would support Cars4U in distinguishing undervalued used cars for acquisition, recognizing overpriced listings, understanding contemporary depreciation patterns, and strategizing inventory and margins. Some clear examples of these supportive advantages include implementing

dynamic pricing, refining inventory, and detecting fraud. Lastly, the Random Forest Regression model scales with the Cars4U business and provides a logical foundation. To reiterate, the Random Forest Regression model can support Cars4U's pricing automation, inventory enhancement, and comprehensive competitive tactics. Chiefly, as the Cars4U business grows, the Random Forest Regression model will continue to perform without much-needed manual intervention.

Recommendations for Implementation

Key Recommendations to Implement the Solution

Cars4U should implement the Random Forest Regression model to serve as the main engine of its pricing workflow. As previously stated, the Random Forest Regression model achieved strong performance metrics: an R-squared value of 0.90 on the test dataset, an RMSE of 3.59, an MAE of 0.88, and an MAPE of approximately 9%. Based on these performance metrics results, the Random Forest Regression model is the most trustworthy candidate for real-world deployment.

To put the Random Forest Regression model into practice, Cars4U should follow the four key recommendations for its implementation. First, Cars4U should incorporate the model into the pricing pipeline. Namely, Cars4U should insert the Random Forest Regression model into the seller onboarding workflow so that every new used-car listing has a computerized, data-driven price. Also, Cars4U should use the Random Forest Regression model's predictions to mark unreasonably priced used vehicles (i.e., overpriced or underpriced) before they are officially listed in Cars4U's inventory. Comparatively, Cars4U should provide its internal team with dashboards that display predicted used-car prices, confidence intervals, and the most influential variables (features). Second, Cars4U should develop a real-time valuation application program interface (API) that exposes the Random Forest Regression model. In detail, the internal API would enable the procurement, pricing, and marketing teams to submit instant valuation requests. Similarly, the model could anticipate future data requests (i.e., cache predictions) for frequently queried used-car listings to reduce computational costs. Third, Cars4U should establish a structured data-quality pipeline to address pervasive missingness in the independent variables New Price, Seats, Mileage, Engine, and Power, as well as in the target variable Price. To clarify, Cars4U should mandate data fields for sellers, verify mileage, engine displacement, and price

ranges when listing used cars, and instantly flag impossible values. Lastly, Cars4U should establish a model-monitoring and training-schedule paradigm for the Random Forest Regression model. To emphasize, Cars4U should retrain the Random Forest Regression model at least quarterly, or whenever a drift is detected. Within the same scope, employees should consistently monitor the RMSE, the MAE, the MAPE, and changes in the importance of independent variables (features) over time. Likewise, employees should sustain a reproducible versioning system for the Random Forest Regression model.

Key Actionables for Stakeholders

Key actionables are listed for each Cars4U stakeholder. For the Data Science Team stakeholders, these stakeholders must deploy the Random Forest Regression model into production, launch automated retraining pipelines, and retain feature-importance dashboards to illustrate valuations. For the Product and Engineering Teams, these stakeholders must implement the pricing application programming interface (API) into the Cars4U platform, fabricate User Interface (UI) elements that display model-generated prices and justifications, and integrate guardrails to detect suspicious listings as part of fraud detection procedures. For the Business and Operations Teams, these stakeholders should use predictions from the Random Forest Regression model to guide procurement decisions, identify undervalued used cars for acquisition, and adapt marketing strategies to accurate location-based price patterns. For Sellers and Buyers, these stakeholders should be provided with clear explanations of price drivers based on variables such as power, depreciation amount, and brand, along with confidence intervals for the price ranges from the Random Forest Regression model.

Expected Benefits and Costs

By deploying the Random Forest Regression model, the following benefits are achieved, as indicated by its performance. In the first place, the Random Forest Regression model offers higher pricing accuracy. In particular, the model has a high test MAPE of 9%, implying that Cars4U can reduce mispricing errors by approximately 20-30%. Second, the Random Forest Regression model yields faster inventory turnover. Specifically, more accurate pricing can reduce idleness; even a 10% improvement in pricing accuracy can lower recurring out-of-pocket expenses by 5-8%.

Third, the Random Forest Regression model can enhance trust and clarity. Explicitly, the model provides feature-importance explanations for independent variables, such as Power and Depreciation Amount, to help defend valuations. Fourth, the Random Forest Regression model can improve procurement decision-making. In other words, the model can identify undervalued cars, increasing margins by 3-7% per used car. Lastly, as earlier noted, the Random Forest Regression Model can aid in fraud detection. In clearer terms, the model can help flag listings that deviate notably from its predictions.

Even though the Random Forest Regression model has notable benefits, the following cost trade-offs must be noted. First, the Random Forest Regression model incurs significant deployment costs. To explain, wherever the model is hosted and running, predictions incur higher infrastructure computational costs due to longer training times. Second, the Random Forest Regression model incurs high data maintenance procedures. To emphasize, the model requires clean, complete data, necessitating frequent operational investments and constant, costly hyperparameter tuning. Third, the Random Forest Regression model requires constant monitoring and retraining. In fact, the model has longer monitoring and retraining times than other models, making it ineffective for quarterly retraining cycles. Lastly, the Random Forest Regression model has higher integration costs. Specifically, the engineering effort required to integrate the model into Cars4U's platform leads to higher memory usage and a considerably larger footprint, which can be an issue on some resource-limited systems or devices.

Key Risks and Challenges

The following are the key risks and challenges highlighted for the proposed Random Forest Regression model. To start with, the Random Forest Regression model cannot operate optimally due to data quality issues in the dataset. To enumerate, the provided dataset had severe missing values and outliers, which can compromise model performance over time. Secondly, the constantly changing conditions of the Indian used-car market can lead to negative model drift or model decay in the Random Forest Regression model. In fact, this means that the model can experience a decline in predictive accuracy over time as Indian used-car market conditions change, including fuel prices, brand popularity, and depreciation patterns. Therefore, the model requires

periodic retraining, and, as previously stated, retraining a Random Forest Regression model incurs higher computational costs. Thirdly, the Random Forest Regression model requires more computational resources, closer monitoring, and more careful version control. In clearer terms, this type of model is bulkier than linear and geometric regression models. Fourthly, the Random Forest Regression model has interpretability limitations, even though it provides feature importance scores. In general, this type of model is less interpretable than linear and geometric regression models. Lastly, the Random Forest Regression model could lead Cars4U's stakeholders to become overly reliant on its performance. To address this overreliance, human oversight remains necessary to ensure that rare vehicles, vintage or modified used cars, and edge-case listings receive proper valuations.

Further Analysis Needed and Associated Problems to Solve

The following are five specific further analyses and associated problems that need to be solved for Cars4U. First, Cars4U should expand the dataset to include additional independent variables (features), such as service history, accident records, seller reputation, and market demand indicators. By doing so, the final Random Forest Regression model would significantly improve its predictive capability. Second, Cars4U should develop a depreciation-forecasting module. By implementing this module, Cars4U can better forecast future values to inform procurement and pricing strategies, as depreciation is a principal predictor of used-car prices. Third, Cars4U can explore alternative regression models, such as XGBoost, LightGBM, or CatBoost, which use gradient boosting and may outperform the final chosen Random Forest Regression model with proper tuning. Fourth, Cars4U can build city-specific pricing models for various Indian cities. To explain, there are strong Indian-city variations in the dataset, where city-specific pricing models can enhance predictive accuracy for used-car prices. Finally, Cars4U should construct a fraud-detection subsystem. To illustrate, Cars4U should use anomaly detection methods to catch suspiciously low-used-car prices, improbable mileage, and erratic engine-power combinations. In summary, by following these implementation recommendations, Cars4U is fully ready to deploy the Random Forest Regression model as a business-ready pricing engine.

Appendix

Baseline vs. Final Regression Model Comparison Tables

Table A1. Performance Metrics of the Baseline Linear Regression Model (originally with too much noise)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.51	0.31	0.50	0.29	7.79	9.27	1.99	2.01	18%	18%

Table A2. Performance Metrics of the Final Linear Regression Model (with multicollinearity and p-values addressed)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.45	0.20	0.45	0.19	8.29	9.97	2.06	2.11	18%	18%

Table A3. Performance Metrics of the Baseline Ridge Regression Model (default parameters)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.51	0.31	0.50	0.29	7.78	9.25	1.99	2.01	18%	18%

Table A4. Performance Metrics of the Final Ridge Regression Model (addressed INF multicollinearity)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.51	0.31	0.51	0.29	7.78	9.26	1.99	2.01	18%	18%

Table A5. Performance Metrics of the Baseline Lasso Regression Model (fits worse than a horizontal line at the mean of the target)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
-0.08	-0.10	-0.09	-0.13	11.63	11.69	5.98	6.21	80%	80%

Table A6. Performance Metrics of the Final Lasso Regression Model (with alpha tuned = 0.1)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.52	0.65	0.52	0.64	7.72	6.61	2.94	2.91	29%	28%

Table A7. Performance Metrics of the Baseline Decision Tree Regression Model (severely overfitted, memorizes training data, and lacks generalization)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
1.0	0.91	1.0	0.91	0.0	3.43	0.0	0.38	0%	2%

Table A8. Performance Metrics of the Final Decision Tree Regression Model (hyperparameter tuned, max_depth = 5, min_samples_leaf = 5, random_state = 1)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.96	0.86	0.96	0.86	2.29	4.20	1.05	1.21	12%	13%

Table A9. Performance Metrics of the Baseline Random Forest Regression Model (too tightly fitted the training data, not learning a stable underlying pattern)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
1.0	0.92	1.0	0.92	0.67	3.13	0.09	0.26	1%	1%

Table A10. Performance Metrics of the Final Random Forest Regression Model (hyperparameter tuned, max_depth = 5, min_samples_leaf = 1, criterion = squared error, random_state = 1)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.97	0.90	0.97	0.90	2.07	3.59	0.79	0.88	9%	9%

Table A11. Performance Metrics of the Baseline KNN Regression Model (with k = 10 neighbors)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.85	0.75	0.85	0.74	4.32	5.53	2.18	2.52	25%	27%

Table A12. Performance Metrics of the Final KNN Regression Model (originally baseline with default parameters)

Train_R^2	Test_R^2	Train_Adj_R^2	Test_Adj_R^2	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	Train_MAPE(%)	Test_MAPE(%)
0.89	0.80	0.89	0.80	3.66	5.04	1.79	2.29	20%	25%